

Stock Synthesis Application to Black Sea Bass

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1. Abstract

Stock Synthesis (SS) was applied to fishery and survey data as recommended by the Black sea bass Research Track Stock Assessment Working Group, to explore model dynamics and provide estimates of stock status. The SS models were developed from the application used in a working paper for the SAW62. The modeled black sea bass population was structured spatially (north and south of a boundary near Hudson Canyon) and seasonally (two semesters), and demographically by age and sex, with hermaphroditism, area-specific recruitment and seasonal fish movement. Recruitment was linked to a time series of winter bottom temperature for the region. Commercial and recreational fishing fleets were modeled by semester and area. The models were fit to catches, discard estimates, survey and recreational fishery abundance indices, retained and discarded length compositions by fishing fleet, age compositions from model-based survey estimates, age-at-length data from fishery and survey sampling, and tag recapture data. Models were able to fit the available fishery and survey data relatively well, particularly with addition of some time-varying components in selectivity and retention to reflect changes in gear and regulations, and recent apparent decreases in size at age. Results are consistent with those of other assessments, estimating that the mid-2010s increase in the stock in the north has slowed, though current abundance is well above management targets. Including the temperature covariate on recruitment improved model fit, corroborating previous research on a positive relationship between winter bottom temperature and survival during sea bass first overwintering period. SS contains some features not available to other modeling platforms applied to black sea bass. Sensitivity analyses to test the consequences for model fits and results to various assumptions were conducted.

2. Introduction

A Stock Synthesis (SS) model for the Northern US stock of black sea bass was developed during the 2016 benchmark assessment, reviewed in SAW62, as part of MAFMC-funded research to investigate alternative assessment frameworks for black sea bass that could help understand implications of spatial structure and other aspects of black sea bass dynamics (e.g. protogynous hermaphroditic life history) and address concerns about the previous benchmark during SARC53 (NEFSC 2017, Fay et al. 2016). The SS application was not adopted for management use, but contributed to the successful review for black sea bass. Simulation testing with various complexities of SS applications demonstrated the need for black sea bass assessment models to account for spatially varying recruitment trends (SAW62 analyses determined that data separated north and south of Hudson canyon provided an appropriate demarcation) to reflect differences in the relative strength of year classes, particularly since 2010 with apparent increases in population

in the northern part of the stock area (Fay and Cadrin 2016, Cadrin et al. 2019). These simulation experiments also demonstrated that including movement among areas was less critical, and results of models that treated northern and southern areas as independent stocks had, at the time, as little bias and comparable retrospective patterns to a single model that included spatially structured dynamics and movement among areas (Fay and Cadrin 2016, NEFSC 2017).

Here we revisit and update the SS application originally developed for SAW62, incorporating data decisions and modeling decisions made by the RTWG, and applying the SS models to the available data for black sea bass up to and including 2021.

3. Methods

3.1. Analytical platform

The analytical approach uses Stock Synthesis (SS v3.30.20, Methot and Wetzel 2013). Among other features, SS has the ability to include spatial and seasonal structure in population and fishing dynamics, flexibility in the parameterization of life history characteristics (e.g., hermaphroditism), allows for mirroring of parameters such as selectivity among fishing fleets, and can fit to a range of available data from fishery dependent and fishery independent sources, including size composition, age-at-length, and tag-recapture data as well as magnitude of catch, magnitude of discards, abundance indices, and age composition information. Environmental variables can also be included, either as covariate effects on model parameters or as data that are treated as being related (with uncertainty) to the values of model parameters or derived quantities. A standardized diagnostics and plotting package in R (r4ss, Taylor et al. 2021) allows for evaluation and comparison of model results.

For ease of description, the model detail that follows represents the configuration and parameterization presented in the ‘base case’ version of the model that sensitivity analyses also described are compared to. Discussion of the process that led to arrival at these base-case settings accompanies each section.

3.2. Model spatial and temporal structure

The model consists of a spatially structured seasonal population model. The model contains two areas, north and south, aimed to characterize the population of black sea bass north and south of Hudson canyon. The model has two seasons, spring (season 1), and fall (season 2), each of equal length (6 months duration). Initial year for the model is 1989, with a terminal year of 2022 (last year of data 2021). All fishery dependent and fishery independent data were split to this spatial and seasonal structure. Several time blocks, corresponding to series of years, were employed in model construction and estimation to facilitate inclusion of time-varying parameters for different modeled processes (explained below). To include the desired spatial variation, the population of black sea bass was modeled via two ‘growth patterns’ (sub-sections of the population with the same life history parameters), with GP1 fish recruiting to the northern area and GP2 fish recruiting to the southern area.

The modeled population age range was from 0-15, with 15 being a plus group. Length structure of the population was generally modeled in 2cm bins, though larger length bins (beginning at 48cm) had coarser resolution. The upper size of the final length bin was 70 cm, larger than

observed individuals because it is necessary to appropriately capture the distribution of length at age for the oldest fish, so growth for these fish is calculated correctly when deriving population quantities such as biomass.

3.3. Biology

3.3.1. Recruitment

A single stock of black sea bass was modeled, with a single stock recruit relationship. The recruitment of age 0 fish into the population occurred at the start of season 2, and the annual recruitment is distributed among the two growth patterns (GP1 and GP2). As each growth pattern only recruits to a single area (GP1 to the north, GP2 to the south), this distribution method divides the annual recruitment spatially. The proportion of fish recruiting to the northern area was modeled to vary over time through a random walk so changes in spatial differences in year class strength could be represented. The random walk parameterization allows years for which recruitment is not estimated (i.e. at the end of the time series and during forecasts) to use the more recent values for the distribution proportions.

Recruitment was modeled to follow a Beverton Holt stock recruit relationship with steepness set to 0.95, and standard deviation for the recruitment deviations (σ_R) of 0.7, as assumed for previous assessments. Annual recruitment deviations from the stock recruit relationship were modeled 1987-2019 (the 2020 year class is not observed until 2021, with only a single year of data).

The regional bottom temperature time series developed in Tabandera et al. (2023 Working Paper) was included as a covariate on the expected recruitment via a multiplicative link to the standardized bottom temperature time series (z-score of the time series from 1989-2021). The coefficient for the environmental link parameter was an estimated parameter.

3.3.2. Growth

Males and females were both modeled within each Growth Pattern, with initial sex ratio at recruitment of 63% female, 37% male (Olaf Jensen, pers. Comm; sex ratio the same for each growth pattern).

Size at age was modeled according to a von Bertalanffy growth curve, with separate parameters by growth pattern. The SS VB parameterization grows fish linearly from the first length bin at recruitment to a length at a designated first age, which was set at 0.75 years reflecting the appearance of age 1 fish in the spring surveys and fishery data; and a maximum length for a second designated older age - here set corresponding to the length of age 8 fish rather than parameterized directly as L_∞ . The CV of length at age was assumed to decrease linearly with respect to age. Parameters governing size at age were assumed to be the same for both seasons.

For growth pattern 1 (northern recruits), the length at maximum age, and VB k parameter were estimated separately by sex within the model. The length at initial age and CV of length at age for both initial and maximum ages were estimated, but mirrored between sex (both sexes used the same values for the CV in length at age).

For growth pattern 2 (southern recruits), the length at maximum age, and VB k parameter were estimated separately by sex within the model. The length at initial age and CV of length at age for the initial ages were estimated but mirrored between sex (both sexes used the same values for the CV in initial length at age). The CV in length at maximum age was fixed at 10% for growth pattern 2 and not estimated within the model, as there was insufficient data to avoid infeasible estimates. The value of 10% was based on inspection of the raw age at length data.

For northern fish, there is evidence for a recent reduction in size at age (Figure 1). A time block in VBK and maximum length for both males and females was included, with separate parameters estimated for the time period 2014-2022. Although there is some evidence within the survey age at length data for a similar change in the south (Figure 1), this is less clear and mainly only noticeable in the spring. The inclusion of the VAST survey index age composition (see Section 3.6.2.3. below) results in SS not fitting directly to much of these data. Time varying size at age was therefore not included for growth pattern 2 (southern recruiting fish).

Time-varying weight-at-length for males and females for each growth pattern was modeled, as fixed inputs based on decadal blocks (1989-1999, 2000-2010, 2011-2022), as recommended by the RTWG (see figures in SS html output). Seasonal variation in the weight-at-length parameters by sex was included, though SS parameterization requires the same multipliers to the weight-at-length parameters be applied for all growth patterns. The difference between spring and fall weight-at-length parameters for both males and females by semester and area was averaged to derive the seasonal adjustments used for both sexes.

3.3.3. Maturity and fecundity

Separate logistic maturity at length relationships were modeled for each growth pattern. The values for the parameters of the maturity at size were fixed inputs based on the information assembled by the WG, and included the same time block changes as the weight-at-length parameters (i.e. decadal changes in parameters). Resulting curves are shown in the SS html output. Spawning output for mature fish was assumed equal to biomass.

3.3.4. Hermaphroditism

Hermaphroditism was modeled explicitly according to sex change rate parameters at age determined from work by Olaf Jensen (pers. Comm., sensu Fay et al. 2016). At recruitment, 63% of fish are assumed to be female. Inflection age for the rate of sex change was set at 4, with standard deviation of age 2 (governing the slope of the relationship with age). The asymptotic rate of sex change was either estimated within the model, or fixed at 0.1. The latter was derived to ensure the emergent sex ratio at older ages / larger sizes was consistent with observed sex ratios. There is in theory information in sex ratio from the age at length data that can inform the maximum rate of sex change. The rate of sex change was not assumed to vary over time.

3.3.5. Movement

Movement between areas is modeled to represent offshore mixing during the winter. At the end of season 2, an estimated fraction of fish in growth pattern 1 in the north move from the North to the Southern area. At the end of season 1, an estimated fraction of the growth pattern 1 fish in the south move from the South to the North. This rate is ostensibly very high to ‘force’ fish back to

their natal inshore area, however there have been some recaptures of north released fish in south inshore region and so this can be non-zero. The two movement rates are modeled as being constant for all years and ages. SS has the flexibility to model age-varying movement, as well as tying movement rates to environmental covariates, but this was not explored here.

3.4. Fishing fleets

Eight fishing fleets were modeled to accommodate gear, spatial, and seasonal differences in availability. Commercial (all gears) and Recreational fishing fleets were thus modeled by semester/season (1 / 2) and area (north / south). This is different from the SAW62 2016 SS application that considered Trawl and Non-trawl commercial fleets separately. Aggregation of trawl and non-trawl fleets was done based on RTWG decisions and to be consistent with the WHAM modeling in this RT. Additional runs were done that maintained disaggregation of commercial fishing fleets to trawl and non-trawl. Some of the need to include time-varying selectivity and retention for the commercial fleets is related to changing proportions of trawl and non-trawl over time, including in the representation of samples from each in the available data.

Selectivity for each fleet was modeled separately as logistic (flat-top) functions of length, with both parameters being estimated. Retention at length was also assumed to follow a logistic curve for each fleet, with parameters estimated. Discard mortality was assumed to be fixed at 100% for the commercial fleets, and set at 15% for recreational fleets (recreational discard estimates fit to in the model include both live and dead discards, Section 3.6.1.2). Catches for commercial fleets are evaluated in biomass, whereas those for the recreational fleets are in numbers. Some degree of time variation in selectivity and retention was required for fishing fleets, to match with changes in regulations and as a result of evident patterning in the data from initial model fits. In all cases these were implemented as time blocked changes in model parameters (with separate estimation of parameters by time block). These time-varying changes were:

- North Commercial Spring: changes in selectivity and retention in 1998, separate asymptote in retention for 2014-2019 to account for change in discard ratio.
- South Commercial Spring: changes in selectivity and retention in 1998, retention asymptote changes in trawl-only discard years.
- North Commercial Fall: separate asymptote in retention for 2014-2019
- South Commercial Fall: changes in selectivity and retention in 1998, retention asymptote changes in trawl-only discard years.
- North Recreational Spring: 3 block changes in selectivity and retention (1989-2000, 2001-2008, 2009-2021)
- South Recreational Spring: 3 block changes in selectivity and retention (1989-2000, 2001-2008, 2009-2021)
- North Recreational Fall: 3 block changes in selectivity and retention (1989-2000, 2001-2008, 2009-2021)
- North Recreational Fall: 3 block changes in selectivity and retention (1989-2000, 2001-2008, 2009-2021)

The included change to the retention asymptote in the Northern commercial fishing fleets reflects apparent increase in the discard ratio in the north as the large 2011 year class makes its way through the fishery.

The included changes to the retention asymptote in select years in the South commercial fleets reflect years with large (but uncertain) discard estimates. These years (1992, 1996, 2004, 2011, 2019, 2020) represent years where the only contribution to the discard estimate is coming from trawl. Not including these changes to fit these data was considered during model development and is presented as a sensitivity run.

3.5. Surveys

Many fishery independent survey data sets are available for black sea bass, in addition to some abundance indices from fishery dependent data (Truesdell and Curti 2023c Working Paper; Brust et al. 2023 Working Paper). The SS model adopted the integration of fishery independent survey information that was available for this RT through application of the VAST model to survey data (Hansell and Curti 2023 Working Paper), consistent with the proposed modeling approach in the WHAM modeling. Sensitivity analyses investigated the consideration of simultaneous fitting to the individual survey information instead of the model-based integrated indices. The SS model also fit to the Recreational Catch Per Angler (CPA) indices, available for the fall. Data from surveys available for inclusion included the time series of abundance, length composition, age composition (VAST models only), and age-at-length data.

The range of possible surveys for which SS models considered (specifics for the base-case discussed below, base-case survey indices indicated here with *) included (along with the season and area used for predictions):

- *VAST Spring, North
- *VAST Spring, South
- VAST Fall, North
- VAST Fall, South
- *Recreational CPA Fall, North
- *Recreational CPA Fall, South
- Massachusetts Spring Trawl survey, North, Age 1+
- Massachusetts Fall Trawl survey, North, Age 1+
- Rhode Island Spring Trawl survey, North, Age 1+
- Rhode Island Fall Trawl survey, North, Age 1+
- Connecticut Spring Age 1, North
- Connecticut Fall Trawl, North
- New York Spring Age 1, North
- New Jersey Spring Trawl survey, South, Age 1+
- Delaware Spring Age 1, South
- Maryland Spring Age 1, South
- VIMS Spring Age 1 Trawl, South
- NEAMAP Spring Trawl, North
- NEAMAP Spring Trawl, South
- NEAMAP Fall Trawl, North
- NEAMAP Fall Trawl, South

- Albatross Spring Trawl, North
- Albatross Spring Trawl, South
- Bigelow Spring Trawl, North
- Bigelow Spring Trawl, South
- NEFSC Winter Bottom Trawl survey, North
- NEFSC Winter Bottom Trawl survey, South
- Commercial CPUE, North
- Commercial CPUE, South

Size selectivity for most surveys were modeled with a separately estimated logistic selectivity pattern at length. Fall surveys were assumed to not catch age 0 fish by forcing selectivity at age 0 to zero. Some surveys, denoted as Age 1, only capture age 1 fish. These were modeled with fixed selectivity at age of 1 and 0 for all other ages. The models assumed linearity in the relationship between changes in the index and abundance for all surveys.

3.6. Data

This section details the available data for fitting the model. The data sets were built following the principles of taking advantage of flexibility of SS while maintaining consistency with the approaches taken for the age-based modeling (WHAM), and incorporating RTWG feedback. In particular, data sets were assembled to emphasize not double-use / borrow data, and data were not included for year-area-season-gear configurations where sampling was sufficiently incomplete for the needed strata (more details below).

3.6.1. Fishery dependent data

3.6.1.1. Catch

Observations of the retained catch were treated as lognormally distributed around the true value of the retained catch by year. The observations were treated as essentially fixed inputs for commercial fleets (catch estimates assigned a log-standard deviation of 0.01). Retained catch (in numbers) for the recreational fleets were assumed to be more uncertain and used the annual estimates of uncertainty (CV) available directly. An initial equilibrium catch was fit to for each fleet to estimate initial F 's to derive the initial year equilibrium population size, with estimate equal to the average by fleet from 1989-1993, and a log standard deviation of 0.05.

3.6.1.2. Discards

Time series of discard estimates (in mt or numbers) were fit to for each fleet by area and season assuming lognormal observation model. Annual estimates of the magnitude of uncertainty in the discards were included directly. For the commercial discards, CVs were available by area, but not by season; the same CVs were used for both the spring and fall estimates but these differed by area. For the recreational discards, discard estimates included both alive and dead discards, and separate time series of annual CVs were used by area and season. As with the catches, the available CVs were converted to log-standard deviations for input for correct use in the observation model.

3.6.1.4. Abundance indices

Fishery dependent index data included the RecCPA time series for both north and south. These data correspond most closely to the fall season and so semester 2 was used to calculate the predicted values. Indices were treated as lognormally distributed around their predicted values. Available annual estimates of uncertainty were incorporated directly by converting the estimated CVs from the index standardization to log-standard deviations. These model-based CVs are low in magnitude and likely do not adequately capture the true variance associated with these data. An additive variance inflation of 0.15 (roughly equivalent to adding 15% to the CVs) was added to the annual uncertainty estimates for both north and south indices to reflect this (effectively down-weighting these time series). Selectivity at length for the RecCPA indices, required to compute the expected values, was mirrored from the relevant recreational fleet. This was able to be done with consistent treatment of the fishery data with the index data because the selectivity curve for the recreational fishing fleets includes all encountered fish, which are included in the RecCPA CPUE standardizations (Brust et al. 2023 Working Paper).

The Commercial CPUE standardized indices were considered for inclusion in sensitivity analyses.

3.6.1.5. Length composition

Fishery-dependent length data included sex aggregated annual compositions for both retained and discarded fish, by year, season, and area. These composition data were not available for all years, with explanation of data decisions to include these data described below. Multinomial likelihood was assumed for these data, with various approaches for defining the initial input sample size, again described below. An approximation to one iteration of the McAllister-Ianelli method for reweighting length data was employed that generally attempted to weight data from each source so the harmonic mean of the annual effective sample size for a given fleet is equal to the adjusted input sample size. For the commercial length data, the discard length compositions were coded into SS as separate fleets (with mirrored selectivity) so that separate weightings could be applied to the retained and discarded composition data. This allows for a recognition that these data are from different data sources. This was not explored for the recreational fleets.

Annual length compositions of the retained commercial retained catch were included for year/fleet combinations (the fleet designation incorporates the seasonal and regional stratification) when sampling was sufficiently complete among market categories (if length observations were not available for a market category that comprised at least 10% of the catch for that fleet, the composition for that year was not used). Some additional year compositions were removed when the number of trips sampled for length was extremely low (though the market category sampling criterion removed most of these). Annual input sample size for the retained compositions was initially set as the number of trips from which length samples were observed in a given year.

Annual length compositions of the discarded commercial retained catch based on observer data samples were included for year/fleet combinations that did not require borrowing from other seasons/areas when calculating discarded catch at length (Curti et al. 2023a Working Paper). Discarded length compositions appear to be computed based on the raw numbers of fish observed, rather than an expanded composition that accounts for sampling protocol, this could be

revisited in future management tracks. Annual input sample size for the discard length data was initially set as the number of hauls from which length samples were observed in a given year. These data were subsequently down-weighted to have an adjusted input samples size of 10% the initial input sample size.

Annual length compositions for the retained and discarded components of recreational catch were included for year/fleet combinations that did not require borrowing from other seasons/areas when calculating catch at length (Curti et al. 2023a Working Paper). Annual input sample sizes were initially set as the number of intercepts from which length samples were observed in a given year. These data were subsequently down-weighted to have an adjusted input samples size of 20% the initial input sample size.

Length composition data from the study fleet, via the Commercial Fisheries Research Foundation (Verkamp et al. 2023 Working Paper), were included as separate composition data. These data were excluded from the likelihood calculations and so were not fit to, but inclusion of the data in the model files allowed for comparison of the data with the model predictions for these data.

3.6.1.6. Age-at-length data

Fitting to age-at-length data directly potentially improves the ability to estimate growth. Fishery age-at-length data from both retained and discarded components of the catch in the spring and fall were included (disaggregated by sex when available) for years where these data were not included in the age-at-length keys incorporated into determining the VAST survey index data (see Section 3.6.2.1 below). Consequently most of the age-at-length data included in the model were from the fall. Fall samples of fish with sizes less than 14 cm were not included, to remove fish that are likely mostly all of age 0 and the assessment is assuming that observations of age 0 fish in the fall are not reflective of the magnitude of cohort strength.

For age at length and age composition data, the base-case model assumed no ageing error. Sensitivity analyses investigated the use of ageing imprecision matrices (by season) based on NEFSC multiple reads (treated as fixed inputs of the variance in observed age given the true age). The values for these were taken from those used in the 2016 SS application to black sea bass (Fay et al. 2016) and were not updated for this application.

3.6.1.7. Tag recaptures

SS can fit to data from the results of mark-recapture experiments. Tag recapture data were available from releases of tagged black sea bass that were made in 2002, 2003, and 2004 (Moser and Shepherd 2008). SS requires each tag release group be assigned an age at release (for convenience of computation), tag releases were allocated between groups of ages 2, 3, 4, and 5 (most were assigned to age 3) based on the size composition of the releases. No updates to tag group assignment was made since the 2016 application, though the recapture data were re-organized to match the aggregation of commercial fleets in the current application. Tag groups that had fewer than 10 observed recaptures over the time series since release were not included in the likelihood. SS assumes that total recaptures from a tag group in a given year are negative binomially distributed, based on predictions from the retained component of the catches. A second likelihood component assumes these tag recaptures are distributed as multinomial among

fleets. Overdispersion for the negative binomial likelihood component was set to 2 for all Tag release groups, as in the 2016 application (Fay et al. 2016), and reporting rate and initial tag loss parameters were fixed at values reported by Moder and Shepherd (2008). The prediction of the number of tag recaptures based on the retained portion of the catch may be in contradiction with how recaptures are reported from the recreational fishery as anglers may have incentive to report high-reward tags from fish they encounter that they would have discarded. An option to calculate expected recaptures based on the selected fish for some fleets might provide an opportunity to investigate this hypothesis.

3.6.2. Fishery independent data

3.6.2.1. Abundance indices

Survey indices used in the base case included the integrated VAST model-based standardized indices for the spring, in both the north and south. These indices, described in Hansell and Curti (2023 Working Paper), integrate data from many of the available survey time series for black sea bass. Abundance indices were assumed to be lognormally distributed around the expected value, with input annual uncertainty estimates (transformation of CVs to log standard deviations). Changes to survey abundance indices were assumed to be (linearly) proportional to changes in the predicted value (no hyperdepletion or hyperstability).

Sensitivity analyses that also incorporated the fall VAST indices were conducted.

Sensitivity analyses that fit to individual survey time series instead of the VAST indices were also conducted. The first of these models included just the federal science center surveys, where the second also included the state survey indices. A detailed exploration of model behavior when alternative combinations of inclusion of various survey data was not conducted. This was in part because the integrated model-based indices provide a current good practice for including data from various sources in this fashion.

3.6.2.2. Length composition

Fishery-independent length data were not fit to in the base-case, because the only composition information available for the VAST indices was age composition data (the survey length data was used to derive the survey specific age data that the VAST models were fit to). A concerted effort was made to fit the VAST indices based on size data rather than ages, however in the limited time available it was not possible to get these models converge successfully.

For model runs that fit to individual surveys instead of the VAST indices, sex disaggregated annual compositions for the surveys were fit to. Each survey has an assigned season and area. Multinomial likelihood was assumed for these data, with initial input sample size set to the number of tows from which length samples were taken when possible, and 5% of the number of fish measured when this data were not available. The 5% was the result of an approximation to one iteration of the McAllister-Ianelli method for reweighting length data was employed that generally attempted to weight data from each source so the harmonic mean of the annual effective sample size for a given survey is equal to the adjusted input sample size. The same scalar was applied across surveys in the sensitivity analyses presented here.

3.6.2.3. Age composition

Age compositions for the VAST indices were included. Rather than use all years of the model-based age composition data, only age compositions for years in which the age length key was constructed based on the preferred approach (WP XXX) of using data from the same season, area, and year were used. This avoids as much as possible double-use of data, and frees up other age observations to be included in the model as age-at-length data. Multinomial likelihood was assumed for the VAST age compositions, with initial input annual sample size set to 25. These data were subsequently down-weighted to have an adjusted input samples size of 50% the initial input sample size.

3.6.2.4. Age-at-length data

Survey age-at-length data from both spring and fall were included, and disaggregated by sex when available (multiple compositions per year for a given length bin are therefore potentially available, corresponding to unassigned sexed fish and the sex-specific composition). As described above in 3.6.2.3, Data were only included for year/season/area combinations where these data were not included in the age-at-length keys incorporated into determining the VAST survey index data. Fall samples of fish with sizes less than 14 cm were not included, to remove fish that are likely mostly all of age 0 and the assessment is assuming that observations of age 0 fish in the fall are not reflective of the magnitude of cohort strength.

For age at length and age composition data, the base-case model assumed no ageing error. Sensitivity analyses investigated the use of ageing imprecision matrices (by season) based on NEFSC multiple reads (treated as fixed inputs of the variance in observed age given the true age). The values for these were taken from those used in the 2016 SS application to black sea bass (Fay et al. 2016) and were not updated for this application.

3.7. Estimated parameters

The estimated model parameters for the base-case therefore included those describing: size at age (von Bertalanffy growth), movement among areas of growth pattern 1 fish, maximum rate of sex change, annual proportions of recruits allocated to growth pattern 1, annual recruitment deviations, link parameter for the relationship between temperature and recruitment, initial equilibrium fishing mortality rates by fleet, catchabilities for the survey and recreational CPA indices, and selectivity and retention at length parameters for fishing fleets and surveys.

3.8. Sensitivity analyses

Additional sensitivity analyses not mentioned above included runs with

- Rate of natural mortality (M) estimated
- Time varying growth for growth pattern 2 fish
- No environmental link to bottom temperature
- Fall VAST indices included (with changes to available age-at-length data based on the same rationale described above for spring)
- Commercial CPUE indices included
- No VAST indices, fitting to Federal fishery independent survey indices and length data (age-at-length data used up in the spring VAST index standardization re-included)

- No VAST indices, fitting to State and Federal fishery independent survey indices and length data.
- Overdispersion in the tag recaptures fixed at 1 (Poisson)

3.9. Reference point calculation, projections

Biological reference points were calculated based on F_{MSY} proxy of $F_{40\%}$, and the average recruitment from 2000-2019, using the average over the last five years of the model growth, selectivity, and fishing pattern among fleets. The SSB reference point (management target) was defined as the equilibrium SSB when fishing at $F_{40\%}$, $SSB(F_{40})$. Projections assumed fishing at $F_{40\%}$ given the average values for growth, selectivity, and fishing pattern among fleets over the terminal five years of the model; recruitment distribution to growth patterns for forecast years followed that in the terminal year (2021).

4. Results and Discussion

4.1. Model estimates, fits, and diagnostics

The model generally was able to fit the available data well, with some exceptions. A full set of diagnostic plots for the base-case SS model can be found in `_SS_output.html` in the repository: https://github.com/gavin fay/bsb-2023rt-ss/tree/main/results/run_279/plots

4.1.1. Growth, sex ratio, and movement

Estimated von Bertalanffy growth parameters resulted in smaller sizes at age for growth pattern 2 (southern recruiting fish), and larger sizes at age for males than females (for both growth patterns), Figure 2. Time blocked estimation of growth for growth pattern 1 resulted in smaller sizes at age in the most recent years, reflecting the observed changes in the raw size at age data (Figure 3).

The estimated asymptotic rate of sex change was 7% per year, resulting in a sex ratio of females:males at the oldest age classes of roughly 20:80 (Figure 4), consistent with observations.

Movement rates for GP1 (northern recruiting) fish were estimated to be 5% from north to the south at the end of season 2, and 85% from the south to the north at the end of season 1.

4.1.2. Selectivity and Retention

Time varying estimates of selectivity and retention for fishing fleets reflected changes in regulations, with decreased retention and selectivity of smaller fish over time (retention and selectivity curves shift to right); Figures 5 and 6. Changes in asymptotic retention for Northern commercial fleets indicates overall higher discard rates for 2014-2019; with lower overall retention also estimated for Southern commercial fleets in the blocked years. These changes indicate that discarding of black sea bass is not wholly size-based. The southern estimated changes in asymptotic retention are somewhat extreme for some years, and may be better represented by penalized deviations, or a constant value for the blocked years rather than annual estimates - this would better reflect the intent of these changes (years for which the discards are solely from trawl). This additional complexity tries to ensure fits to the discard estimates, though

it is noted the estimates for these years have high relative CVs (see Section 4.1.4 below). Disaggregating commercial fleets by gear could allow for better representing changes in discard behavior associated with changes in the proportional makeup of catch, though in 2016 this did not completely resolve such patterns; commercial fleets were aggregated here to both make better use of available data and compare with age-based assessment approaches in WHAM. Estimates suggest that only time-varying retention in the south may be sufficient (i.e. not estimate changes in selectivity, Figure 6c, 6g).

4.1.3. Catch

The time series of catches, as well as the estimated equilibrium catches, are fit to almost exactly, which is unsurprising as these are essentially treated as fixed inputs.

4.1.4. Discards

Estimates of the magnitude of discards largely mirrored trends in the observations, with most estimates being within or close to the input uncertainty ranges for these data (Figures 7 and 8). Some departure from these fits are seen in the recreational fishery in the south, particularly in the fall. Imposed changes over time in the asymptotic retention rate discussed above were required to be able to capture the recent increases in discards in the north, and also the occasional years with high discarding in the south. One exception is the predicted discards in the southern commercial fleet in the fall, with the model estimate of 5,611 mt compared to the observation of 167 mt. However, this high estimated discard is consistent with the observed discards in the recreational fishery in the south in the same year.

4.1.5. Indices of abundance

The time series of spring VAST survey indices are fit well (Figures 9 and 10), though the variability in the index data in the south is not captured as well as the north. The magnitude of the increase in the survey index in the south during the early 2000s is under-estimated, and some of the low index values in the south tend to be over-estimated, which can be seen in the diagnostic plot of observed vs expected values (Figure 10). Fits to data in the 2016 SS application indicated there was evidence in some surveys for non-proportionality in the relationship between biomass and survey index values (e.g. hyperstability and/or hyperdepletion), that may be contributing to the results here though model runs that estimated a power parameter in the prediction of survey index values were not conducted for this analysis.

Fits to the recreational CPA indices capture the general trend for these time series, including the sharp increase in the north after 2013 (Figures 11 and 12). There is some patterning in residuals over the time series, particularly in the south (Figure 12).

4.1.6. Length composition data

The fishery length composition data are overall fit well (Figure 13), with the changes in length frequency in both the retained and discarded information over time being accounted for by both variability in year class strength, as well as the blocked time variation in the retention and selectivity at length (full plots of annual fits to the length data in the html output). Some residual patterns remain in the fits to the length data however (Figures 14-16), most noticeably in the

recreational data (Figures 15 and 16) which show evidence for either a gradual shift in retention during the early 2000s rather than a step-wise annual change, or a failure during this time period to capture cohort strength. Changes in recreational management measures by state over time likely resulted in some variability in fishery response when viewed in aggregate at the regional level. Separate weightings for the commercial retained and discarded compositions were needed to prevent the retained composition from being underfit. Improvements in fits to length composition data could likely be made with additional reweighting, whether via the implemented multinomial or via the self-weighting dirichlet-multinomial. Nevertheless, inspection of effective sample size vs input sample size, as well as Francis method mean length plots, showed that the weightings used in the base-case provided a good approximation (see html output for these diagnostics).

The implied fits to the CFRF discarded length compositions (Verkamp et al. 2023 Working Paper) suggest these data are consistent with the model estimates for some years (Figure 17), though there is some variability among years. In particular, study fleet samples in the fall in the north during years 2017-2019 contained discards typically of large (~60cm) fish, which were not the primary part of the observer samples for those years.

4.1.7. Age composition data

The VAST survey index age composition data are largely fit well (Figure 18), though the 2011 year class is under-estimated after age 2 (Fig. 18a) and there is some under-estimation of the 2000, 2015, and 2016 year-classes in the south (Fig. 18b).

4.1.8. Age-at-length data

Estimates are able to capture the trends in the age-at-length data - particularly when including sex-specific and time-varying growth in the model. There is some interannual variability in the age-at-length data that does not appear to be consistent with consistent changes over time, whether by cohort or year, that could reflect availability to sampling. Even with the time block on von Bertalanffy growth parameters that changes size at age for growth pattern 1 fish, the summary of the fit to the most recent years of data (e.g. 2020 and 2021) using mean age at length indicates that sea bass in the north were smaller than predicted. This is observed in both the fishery and survey age-at-length data (Figure 19, 20).

4.1.9. Tag recapture data

The model is able to capture the overall trend in the number of recaptures, including the magnitude of recaptures in the initial years following release, and the general pattern of tag recaptures among fleets. Predictions of total tag recaptures several years following release are greater than the numbers observed, and predicted numbers of recaptures for recreational fishing fleets are less than those observed (Figure 21).

4.2. Model estimates and stock status

4.2.1. Recruitment

The model estimated that the proportion of annual recruits by area has shifted over the time series, from the majority being in the south during the 1990s, to the majority being in the north from 2005 on. The time series of estimated recruitments shows strong year classes in 2011, 2015, 2018, and 2019, with the bulk of these recruits being in the northern area (Figure 22). Unlike the previous 2016 SS application, and the WHAM model estimates during this RTWG, the SS model does not estimate variable and high recruitments in the early part of the time series - for example, the 1998 and 1999 year classes are not overly large, as was evident from the fits to the index data. This is most likely due to the sparseness of composition data during the early part of the time series - there are few years of fishery and survey data in this time period with representative sampling. Sensitivity analysis that fit to individual surveys and their length composition data provided an avenue for increasing the amount of composition information (albeit through lengths) during this earlier time period.

The estimated value for the environmental link coefficient that accounts for the covariate effect of the winter bottom temperature time series was 0.587, indicating a strong positive correlation between the bottom temperature time series and magnitude in recruitment. Consequently, the RMSE of estimated annual recruitment deviations was reduced by ~45% when the bottom temperature link was included. This may suggest that a lower value for σ_R is appropriate, although this variance partitioning is better treated using a marginal likelihood framework. Consequence of this positive relationship is that estimated recruitment deviations are not necessarily as large for some years (e.g. 2011) as is estimated when the covariate is not included, because the environmental relationship results in an expected recruitment for those years higher (or lower) than the stock-recruit relationship (e.g. Figure 22, bottom left panel).

4.2.2. Spawning biomass

Estimated time series of spawning output (Figure 23) shows the large increase in stock size in the north following the 2011 year class. Spawning biomass is estimated to have declined somewhat as this cohort progressed through the age structure and fishery, though the strong year class in 2015 means this decline has not been abrupt and SSB remains well above the biomass associated with equilibrium stock size at the target fishing mortality rate of F40%. The model estimates the bulk of the biomass to be in the north. The estimate of annual movement rates of northern fish to the south in the fall and from the south to the north in the spring were low (11%).

4.2.3. Fishing mortality

Estimates of fully-recruited fishing mortality (ages 4-7) were relatively high (>0.4) in the 1990s, decreased in the 2000s to below 0.4 since the mid-2000s (Figure 24). SPR is estimated to have remained above the target.

4.2.4. Retrospective analysis

Strong retrospective patterns were observed in SSB and F, with moderate pattern for recruitment (Mohn's rho values: SSB = -0.309, Rec = 0.104, F = 0.495), as shown in Figure 25. Estimates of

stock size (e.g. SSB) decreased with increasing peels (years of data removed). This result is not surprising given that post-2010 increases in the stock in the north have been due to a series of large year classes, the estimated magnitude of which decreases with fewer years of data. Additionally, the 2014-2021 part of the time series that is modified during the retrospective analyses is also when the model assumes time-variation in both growth and in the discard rate for commercial fleets in the north. The latter changes the magnitude of removals from the stock - fewer years of data on the change in discard rate likely means the discard data are fit to less well, implying lower removals for the same perceived relative change in abundance, leading to a smaller estimated stock size.

4.3. Sensitivity analyses

A set of sensitivity analyses explored changes to the fit of the model, as well as to key derived quantities, for changes to the base-case (i.e. each sensitivity only changed the focal parameter/assumption, changes were not cumulative over runs). Table 1 summarizes the results of these runs (runs shown do not include the full set of runs conducted during model development, key decision points associated with arriving at base-case settings are discussed above). Sensitivity analyses were generally not optimized (exception mentioned below).

Not including the recruitment linkage to bottom temperature resulted in a poorer overall model fit, fitting the length and age data less well than the base-case as well as not capturing the initial equilibrium catches (Table 1). Not including the link to the environmental time series resulted in a smaller estimated stock size and higher rates of fishing mortality (Table 1, Figure 26).

Fixing the rate of natural mortality at 0.55 yr^{-1} (base-case value 0.4 yr^{-1}) resulted in an improved overall model fit, with better fits to the discard data, length compositions, and number of tag recaptures than the base-case. Slightly poorer fits to the abundance index and age data. The higher M values led to a higher estimated stock size, including higher 2022 SSB relative to the biomass target (Table 1).

Comparable to the WHAM modeling, the base-case model did not include ageing error. Adding ageing error based on information from NEFSC on multiple reads (Fay et al. 2016) resulted in an improved fit to the age data (Table 1). This was not surprising given the high variability in age estimation for age 0 and age 1 fish. There were several observations of age 0 fish in the fall age-at-length data, even when small fish had been removed from the dataset, so adding ageing error allows for less penalty associated with the lack of fit to these data (we assume no age 0s are available to fishing fleets or surveys), as well as the increased uncertainty in true age for older fish. Adding the ageing error resulted in very little difference to the model predictions (SSB, F , etc.; Table 1, Figure 26).

Adding the fall VAST indices and age compositions, which also changed the available fishery and survey age-at-length data, resulted in improved fits to the discard estimates, length compositions, and numbers of tag recaptures (Table 1). This model estimated a smaller absolute stock size than the base-case, a smaller relative increase in the spawning biomass over the modeled time series, but a higher stock size relative to the biomass target reference point (Table 1, Figure 26).

Replacing the model-based (VAST) indices and age compositions with the federal survey indices and their length compositions increased the amount of age-at-length data available, though also increased the number of estimated parameters associated with fitting these data (catchability and selectivity parameters). An initial run suggested that the NEAMAP length data (particularly in fall) could be fit better with dome-shaped selectivity; subsequently all four NEAMAP indices were assigned dome-shaped selectivity. This is consistent with the assumption made in the 2016 SS application (Fay et al. 2016), and meant that the strong cohort apparent in these data (the length data suggest the majority of the survey catch is from the strong year classes during the 2010s) could be fit better, with less patterning in the residuals for these data. Overall, this model did not fit the discards as well as the base-case, though there was an improvement in the fit to the number of tag recaptures (Table 1). Results for model estimated quantities from this run were very similar to the base-case for this model (Table 1, Figure 26).

A model that forced GP1 fish to move from the south to the north at the end of season 1 (i.e. only GP2 fish are in the south during season 2) resulted in an improved fit to the index data, but had a poorer fit to the observations of tag recaptures by fleet (Table 1).

4.4. Reference Points

Total SSB was estimated to have been above the annual SSB ($F_{40\%}$) reference points since 2009 (Figure 23), and the fishing mortality was estimated to have been below the annual $F_{40\%}$ reference point since 2003 (Figure 24).

To develop the reference points used for management, recent black sea bass assessments have used the most recent 5-year average of length-specific maturity, SSB weight, catch weight, fleet selectivity, and natural mortality estimates to calculate $F_{40\%}$, and the average annual recruitment for years after 1999 to estimate SSB at $F_{40\%}$ (NEFSC 2017). We used as close to these same configurations as possible for the SS model.

The total estimated fishing mortality (average over ages 4-7) that achieved 40% of unfished spawning biomass per recruit was $F_{40\%} = 0.407$ (standard error (SE) = 0.015). The estimated total equilibrium SSB at ($F_{40\%}$) was 19,042 mt (SE = 493 mt). The estimated total equilibrium yield (retained and discards) at $F_{40\%}$ was 6,690 mt (SE = 213 mt). Total stock-wide SSB was estimated to be 217% of the reference point and fishing mortality was estimated to be 72% of the reference point in 2021.

The objective of this research track is to develop the assessment and projection methodology that will be used in subsequent management track assessments. As such, stock status recommendations are not part of the research track Terms of Reference and the results from this research track assessment will not be used directly in management. Instead, this research track assessment will inform a management track assessment scheduled for June 2024. The 2024 management track assessment will provide updated estimates of stock status using data through 2023 and will be used to inform management measures for 2025-2026.

4.5. Projections

To demonstrate the projection methodology and comparison to Multi-WHAM, we configured a three-year projection (2022-2024) with catch set to 10,000 mt in 2022 and fishing at $F_{40\%}$ in the

subsequent two years. Following the methods used to estimate reference points under prevailing conditions (Section 4.4.), average annual recruitment estimates for years after 1999 and the most recent 5-year average of length-specific maturity, SSB weight, fleet selectivity (by fleet), and natural mortality estimates, and fishing pattern among fleets were used to conduct short-term projections. As recreational fleet catches are in numbers whereas commercial fleet catches are in biomass, the distribution percentages of the fixed 2022 dead catch for the recreational fleets were computed using the model-estimated ratio of numbers:biomass in the catch for the most recent 5 years as the basis for obtaining the fishing pattern (in biomass) over all fleets for 2022. Thus the actual projected catch in 2022 differed slightly from the desired input of 10,000 mt (9,634 mt).

When fishing at $F_{40\%}$, the projected SSB is reduced (Table 2). The projected recruitment (94.4 million fish) includes the large (but uncertain) estimated 2019 year class in the calculation of the average. Even with the projected decline in SSB, the stock size in the terminal year of the projection is still far above the corresponding reference point.

4.6. Discussion

The SS applications indicate that the spatial, seasonal, length, age, sex, fleet structured model fits the available data well. Adding time varying discarding rate provides flexibility for the model to fit the discard data, which is relatively uncertain. However, model estimates from time-invariant asymptotic discard rate are similar. The consequences of modeling discards this way is that the model sometimes estimates relatively high fishing mortality in some years resulting in large decreases in biomass that are not immediately apparent in the index data. However, without time varying discards, it is not possible for the model to estimate increases in discards in the north in recent years.

The models clearly reflect trends in the data suggesting the increase in abundance in the northern region as a result of several large year classes during the 2010s (2011 being the highest). This results in an estimate for the stock that is well above management targets in the terminal year, even with recent estimated slight decreases in abundance. Rate of fishing mortality has also remained below the FMSY proxy. The terminal year model estimates are sensitive to the magnitude of information about recent large year classes, and the magnitude of removals, as evidenced by the strong retrospective patterns observed. Unlike other modeling approaches applied to black sea bass, the SS application here did not estimate large recruitments in the south during the late 1990s, this may have been due to the sparsity of data included in the model fitting during this time period.

Inclusion of bottom temperature as a covariate on recruitment resulted in an improved fit to the various data sets, the estimated positive correlation of recruitment with the temperature index was appreciable. Additional model runs could be done applying different temperature time series (e.g. winter temperature in the north only), and linking them to both the overall recruitment or the proportion of recruits in the northern region. Linking overwintering temperature to recruitment success from a conceptual perspective seems most likely to be apparent (or dependent) in the northern region - because of the way SS models recruitment (overall value and then allocation to growth patterns via proportions), it was not possible to link temperature to the number of northern recruits directly. The temperature index was included via the covariate approach. SS also has the capability to treat environmental time series as 'data' informing on the value of an annually varying parameter (such as recruitment deviations). This was not attempted

here but could be included in future applications for including environmental effects in black sea bass assessment models.

Adoption of the model-based indices (VAST output) seems parsimonious - when the individual survey indices were fit to rather than the VAST indices, the overall assessment results were similar. Using the model-based VAST age compositions does mean that an appreciable quantity of age-at-length data, that could be used to inform growth as well as relative cohort strength, is 'used up' in the development of the VAST age compositions. Ideally, the VAST model could be fit to length composition data rather than ages (models requiring age compositions could apply age-length key after obtaining the VAST results, rather than before). Such an analysis was attempted during the RTWG, but it was not possible to obtain converged model results in the time available. A comparison between model-based index and composition results when age length keys are applied prior to or following the analysis would be a useful exercise. Further, it is not clear how to define sample size for these model-based composition data, given they integrate over a large (and varied) number of surveys with different operating protocols and spatial extents, though a move to using a self-weighting likelihood would alleviate such concern.

Simple assumptions were made regarding the timing of surveys and fishery catches (i.e., all take place in the middle of the season). This is a simplification of the actual timing of these data (e.g., Many of the inshore catch in season 1 is caught in the last 2 months of the semester, similarly much of the catch in season 2 is caught early in the semester). These differences in timing could influence the estimation of growth, particularly emphasized due to the seasonal growth pattern whereby age 0 fish in the fall are essentially the same size as age 1s in the spring. This is also the case for older ages but is less pronounced. This issue will warrant consideration in size-based approaches to modeling black sea bass. To reduce the influence of new recruits on the estimation of growth, we removed small fish from fall length and age at length data, and set selectivity for the fall age at length data to 0 for age 0.

Assumptions about biology and life history of black sea bass have impacts on model fit and resulting stock status estimates. Time varying growth resulted in better fits to the data, with recent evidence for decreases in the sizes at age of sea bass. The available data is such that it was not possible to fully estimate growth curve parameters for the south. Models that allowed the rate of natural mortality, M , to be estimated gave higher values than the fixed values used in the base-case run. It was not possible to estimate M for the southern growth pattern, hence the fixed M of 0.55 in the sensitivity analyses. This model generally fitted the data better. The SS model included hermaphroditism, based on current best estimates and fits to data on sex ratio from the available age-at-length information. Sensitivity analyses that investigated the consequences of alternative assumptions for the parameters governing the sex ratio and rate of sex change would likely be a suitable candidate for further model run exploration.

Movement (level of mixing) was assumed to be constant over time, and largely is informed by the tag recapture data from the tag releases in the mid-2000s. There is little direct data from the fishery or surveys to inform time-varying movement, though the extent of this mixing likely changes over time, in part due to changing habitat (e.g. Tabandera et al. 2023 Working Paper). The estimated amount of mixing during the winter (proportion of northern fish moving to the south) here was lower than that estimated in the SAW62 SS application (5% here compared to 11%), and this lower estimate may be informed by changes in the other data streams that could be indicative of fewer fish moving and mixing to the south as temperatures increase. Finding

ways to include proxies for the extent of this mixing, such as suggested in Tabandera et al. (2023 Working Paper), within assessment models might at least allow for a scenario-based approach to understand potential range of implications for changes to these dynamics.

5. References

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6. Tables and Figures

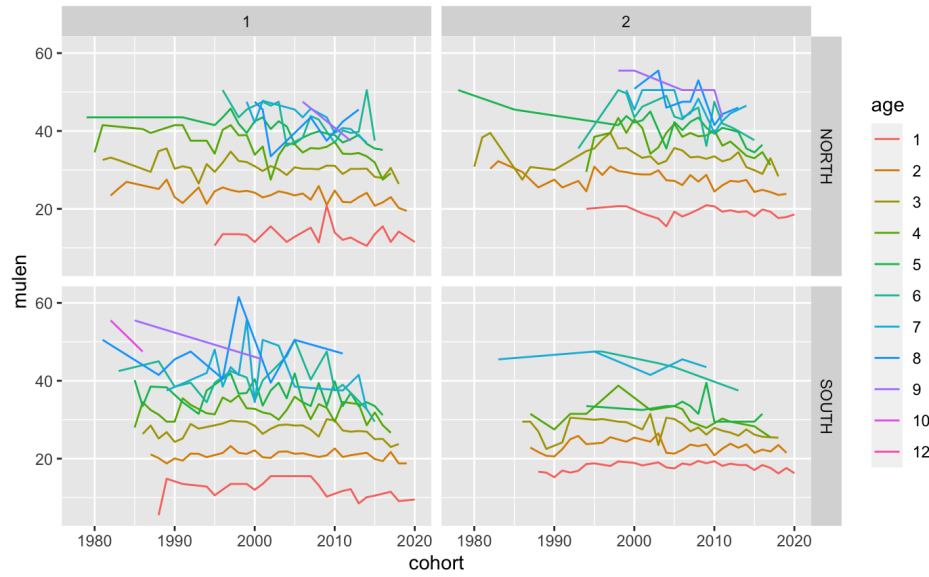
Table 1: Contributions to the objective function and derived quantities for the base-case and set of sensitivity analyses.

	Objective function values (nLL)									Derived Quantities				
Run	Total	Catch	Equil catch	Survey	Discard	Length comp	Age comp	Tag comp	Tag negbin	Recruitment	SSB 2022	F 2021	SSB2022/SSB40	SSB2022/SSB1989
base-case	6851.1	0.1	1.6	-25.0	703.6	2718.0	1843.0	1181.2	421.9	-4.5	41,365	0.254	2.17	6.86
no bottom temperature link	6913.7	0.1	17.9	-15.5	696.8	2734.0	1859.1	1181.2	429.5	-0.2	28,983	0.359	1.64	5.80
M = 0.55	6708.4	0.0	1.3	-18.5	647.6	2683.2	1848.2	1172.0	349.9	13.1	59,033	0.204	3.33	6.67
add ageing error	6831.8	1.3	1.4	-26.1	697.8	2726.7	1805.2	1184.6	432.9	-4.7	41,198	0.271	2.10	6.78
add fall VAST indices	5876.1	0.1	1.1	83.8	653.5	2694.7	845.6	1184.7	403.0	-5.0	31,565	0.263	2.09	6.12
federal survey indices, no VAST	10123.2	0.1	1.9	95.0	822.9	4490.0	3105.4	1203.8	393.9	-3.8	37,975	0.270	2.11	7.30
force move to north	6990.8	0.1	1.7	-34.4	695.3	2726.8	1850.5	1321.7	423.1	-4.7	44,140	0.256	2.22	7.20

Table 2: Projected SSB and total yield under fixed catch for 2022 and F40% for 2023-2024.

Year	SSB (mt)	Yield (mt)
2022	44,462	9,634
2023	40,398	14,517
2024	34,086	12,003

(a) females



(b) males

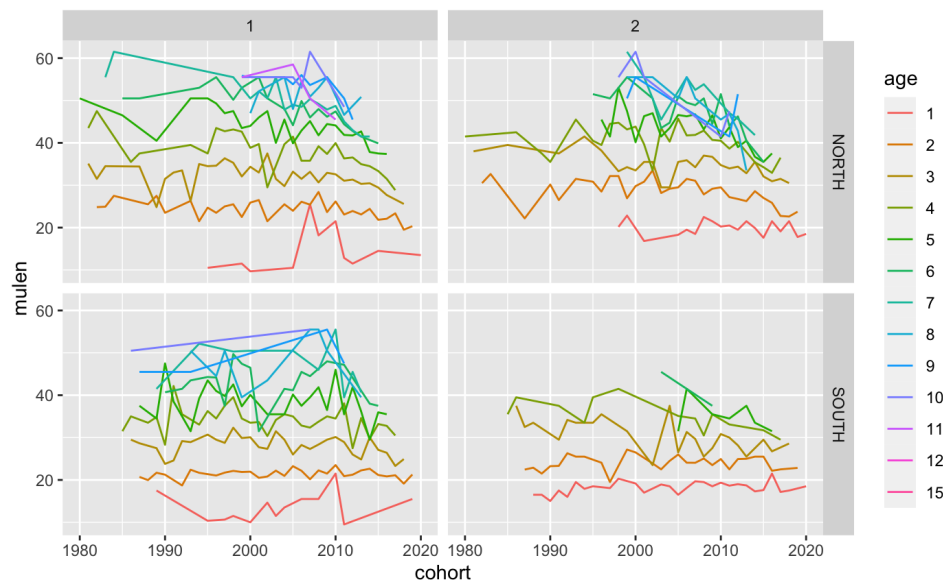
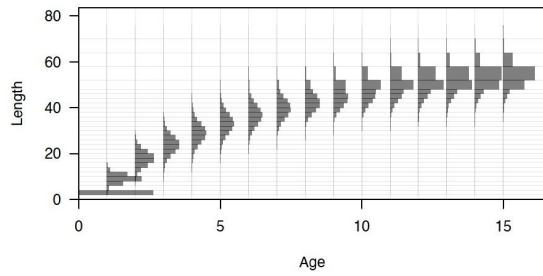
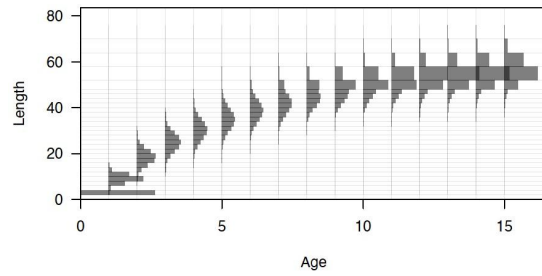


Figure 1. Change in mean size at age by cohort, structured by area (north/south) and semester (1=spring, 2=fall) from the survey age-at-length data for (a) females, and (b) males.

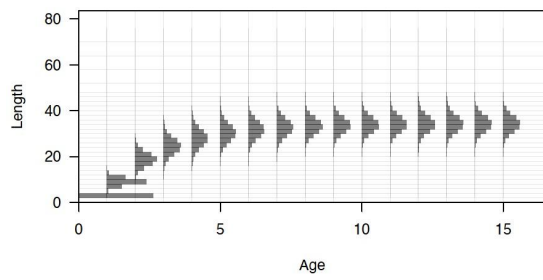
(a) Growth pattern 1, females



(b) Growth pattern 1, males



(c) Growth pattern 2, females



(d) Growth pattern 2, males

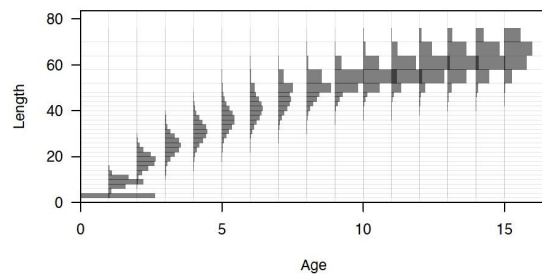
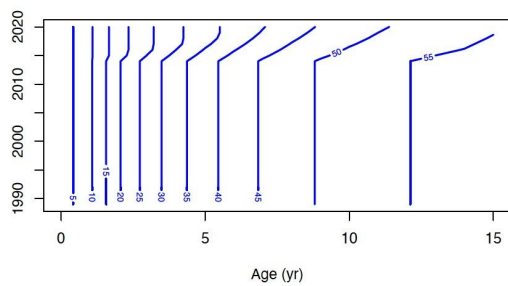


Figure 2: Distribution of length at age in season 1 by growth pattern and sex

(a) Growth pattern 1, females



(b) Growth pattern 1, males

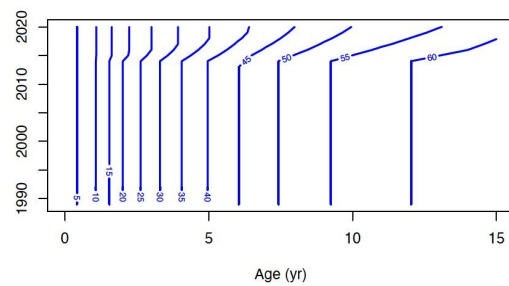


Figure 3: Mean length at age over time for growth pattern 1 by sex.

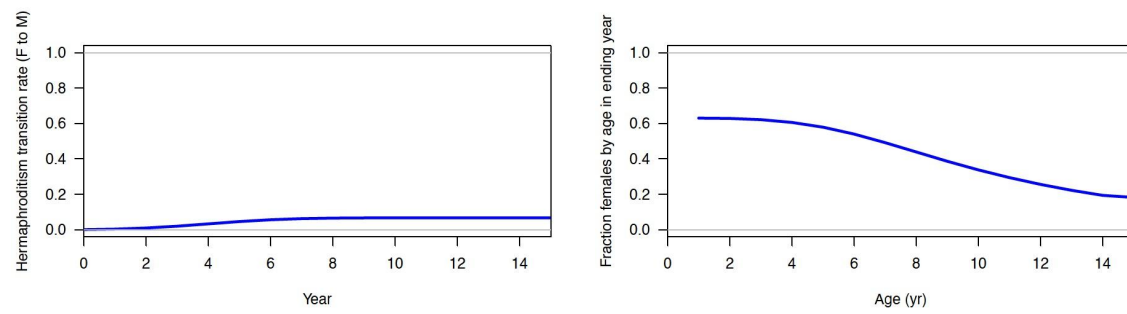
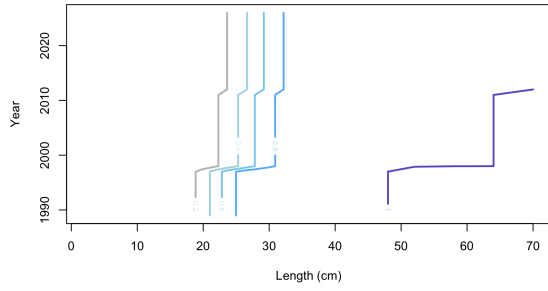
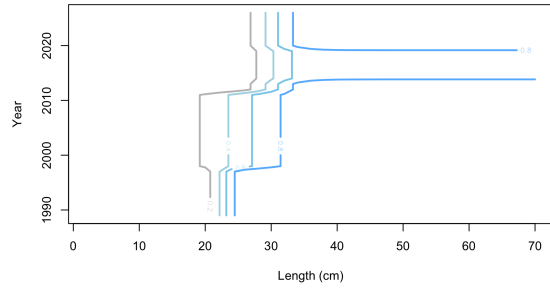


Figure 4: Estimated rate of sex change (left) and sex ratio in the terminal year (right).

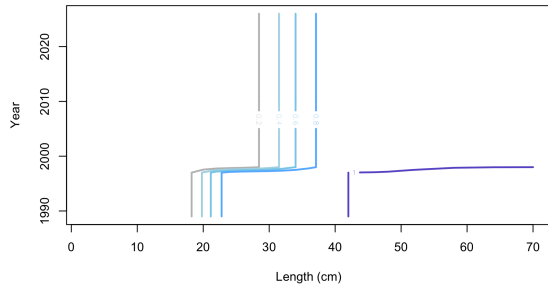
(a) Selectivity at length, N Comm Spring



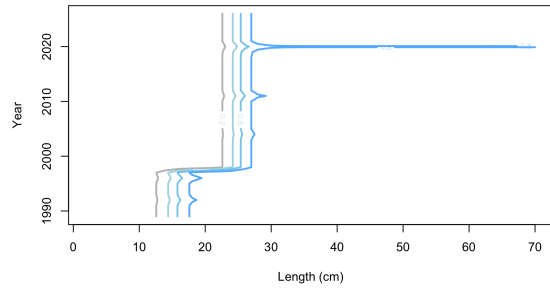
(b) Retention at length, N Comm Spring



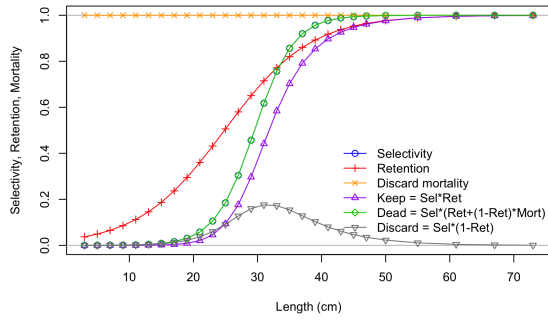
(c) Selectivity at length, S Comm Spring



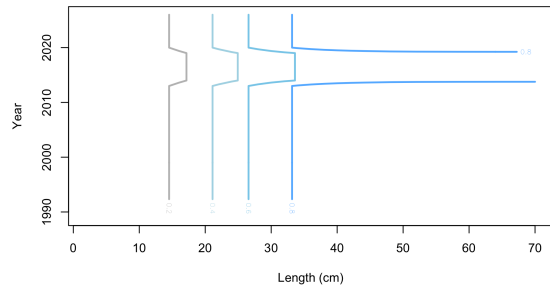
(d) Retention at length, S Comm Spring



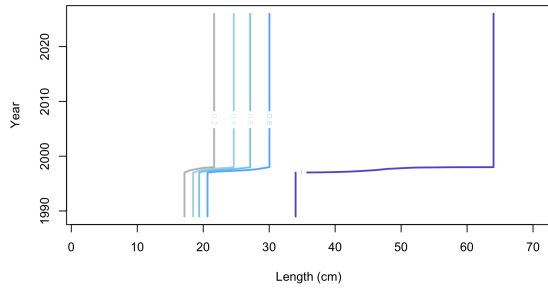
(e) Selectivity at length, N Comm Fall



(f) Retention at length, N Comm Fall



(g) Selectivity at length, S Comm Fall



(h) Retention at length, S Comm Fall

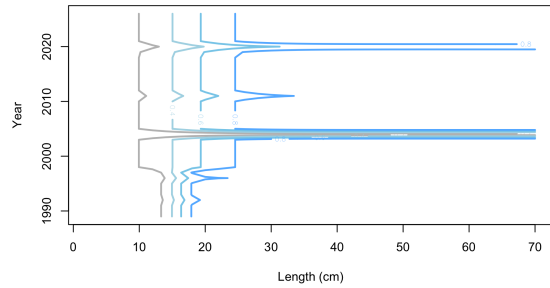
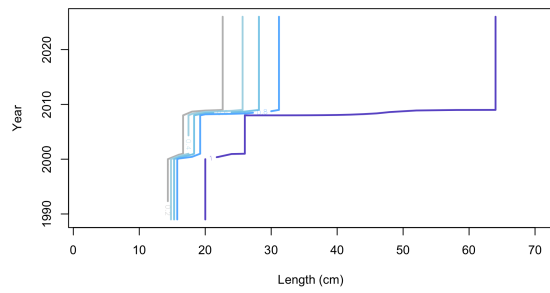
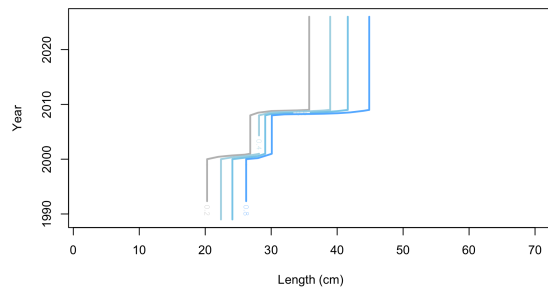


Figure 5: Estimated selectivity and retention at length for commercial fishing fleets

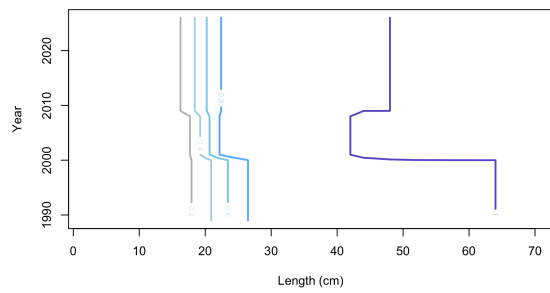
(a) Selectivity at length, N Rec Spring



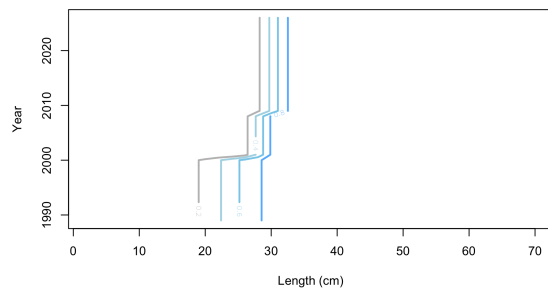
(b) Retention at length, N Rec Spring



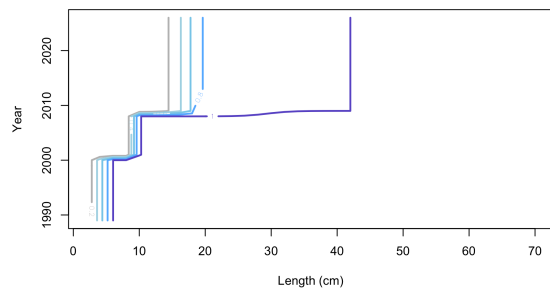
(c) Selectivity at length, S Rec Spring



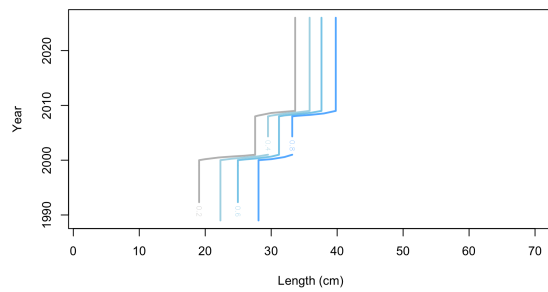
(d) Retention at length, S Rec Spring



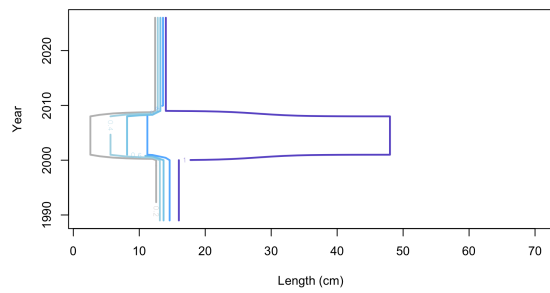
(e) Selectivity at length, N Rec Fall



(f) Retention at length, N Rec Fall



(g) Selectivity at length, S Rec Fall



(h) Retention at length, S Rec Fall

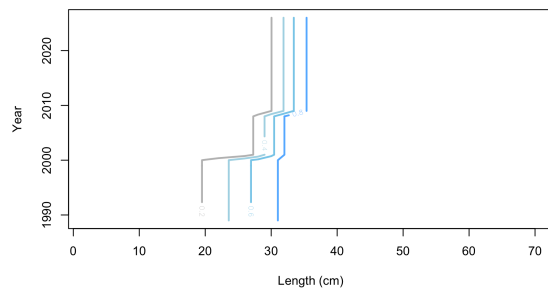


Figure 6: Estimated selectivity and retention at length for recreational fishing fleets

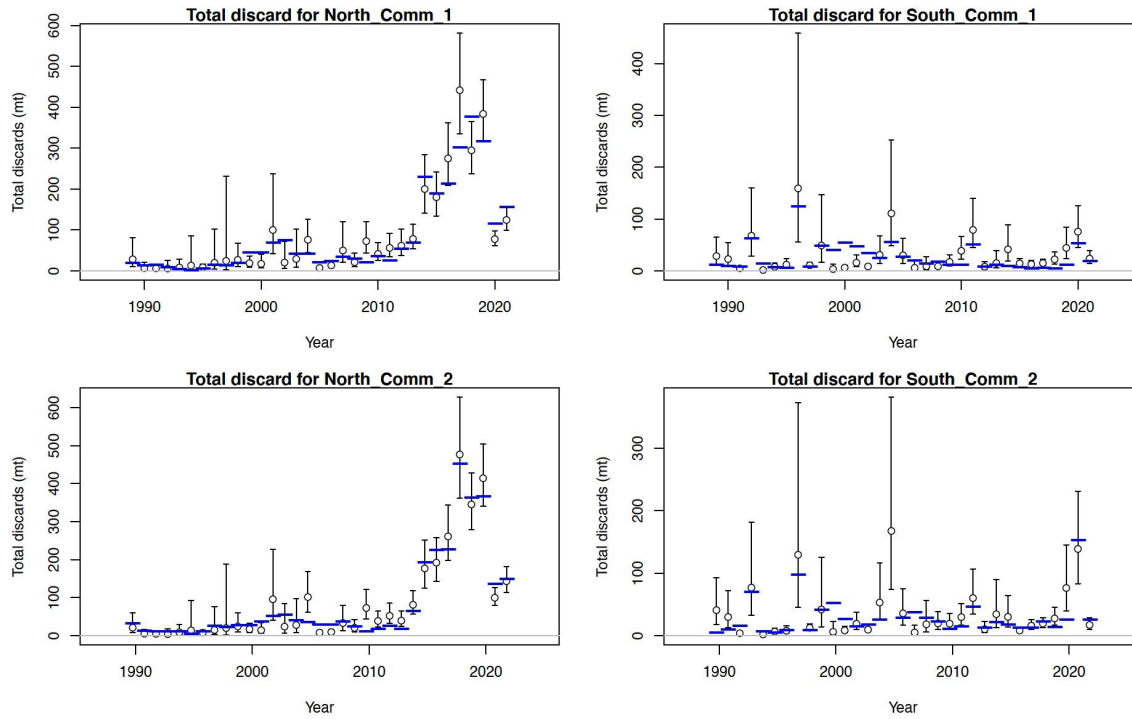


Figure 7: Fits to Commercial discard estimates. Season 1 = Spring, Season 2 = Fall

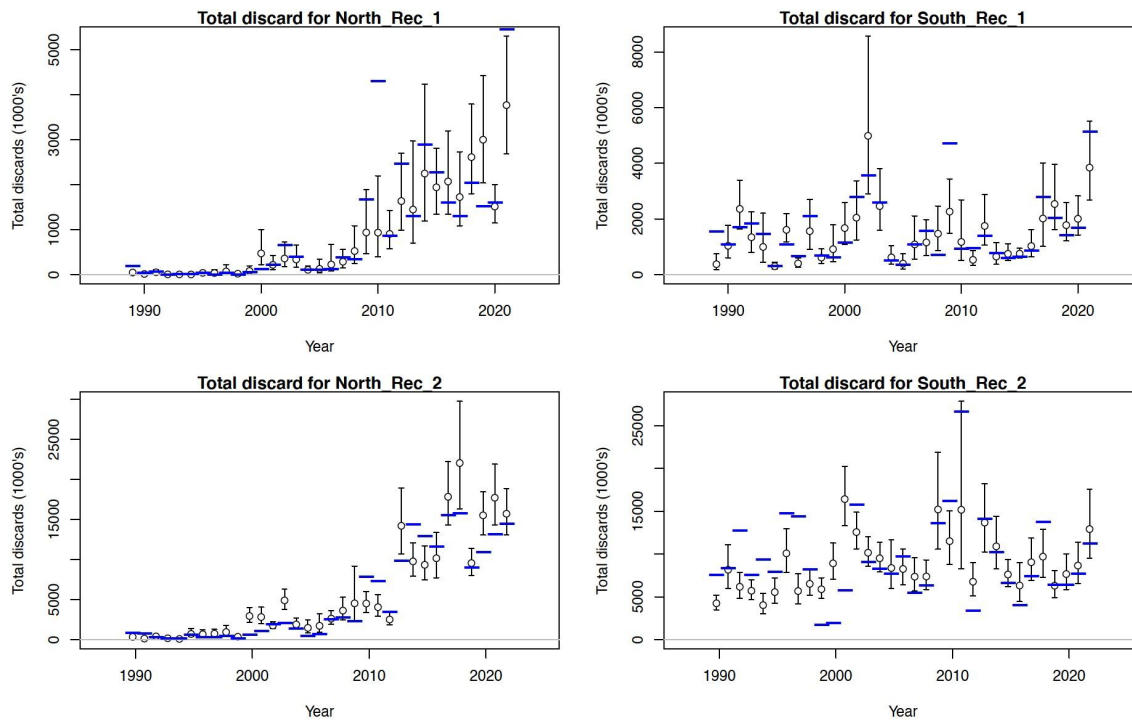


Figure 8: Fits to Recreational discard estimates. Season 1 = Spring, Season 2 = Fall

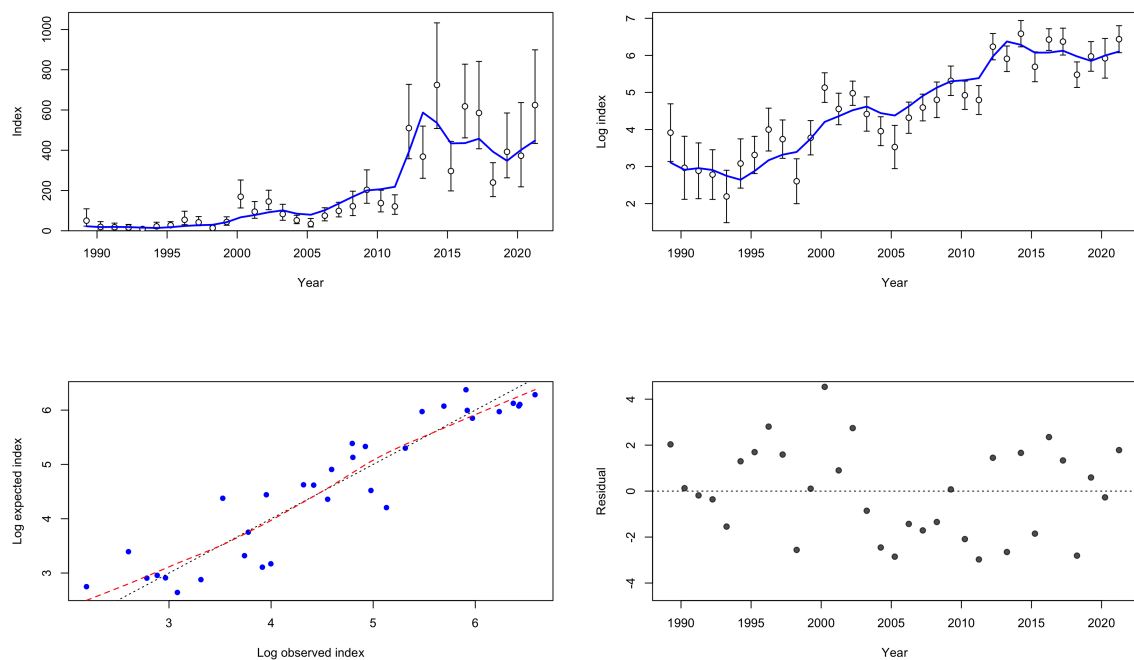


Figure 9: Model fit to Spring North VAST index

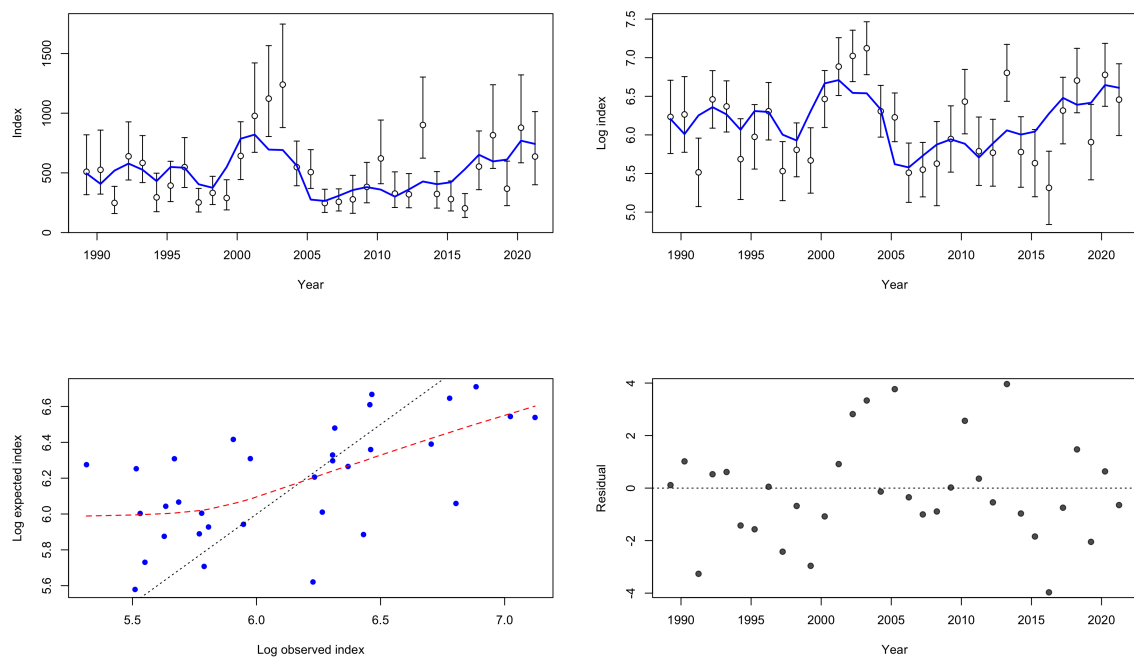


Figure 10: Model fit to Spring South VAST index

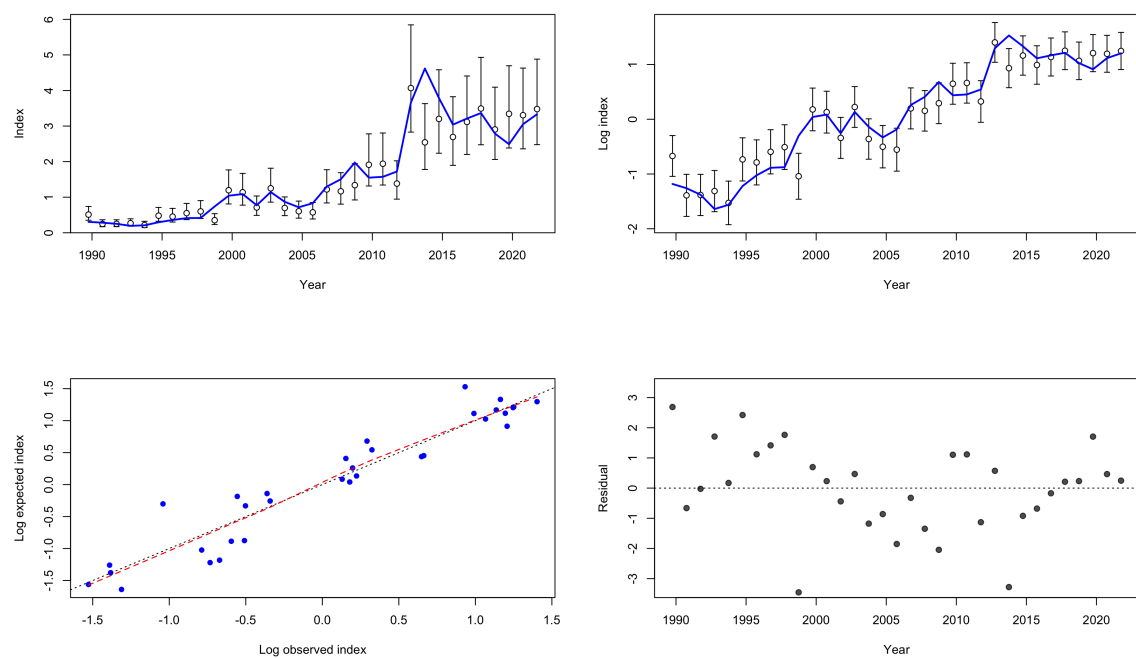


Figure 11: Model fit to North RecCPA index

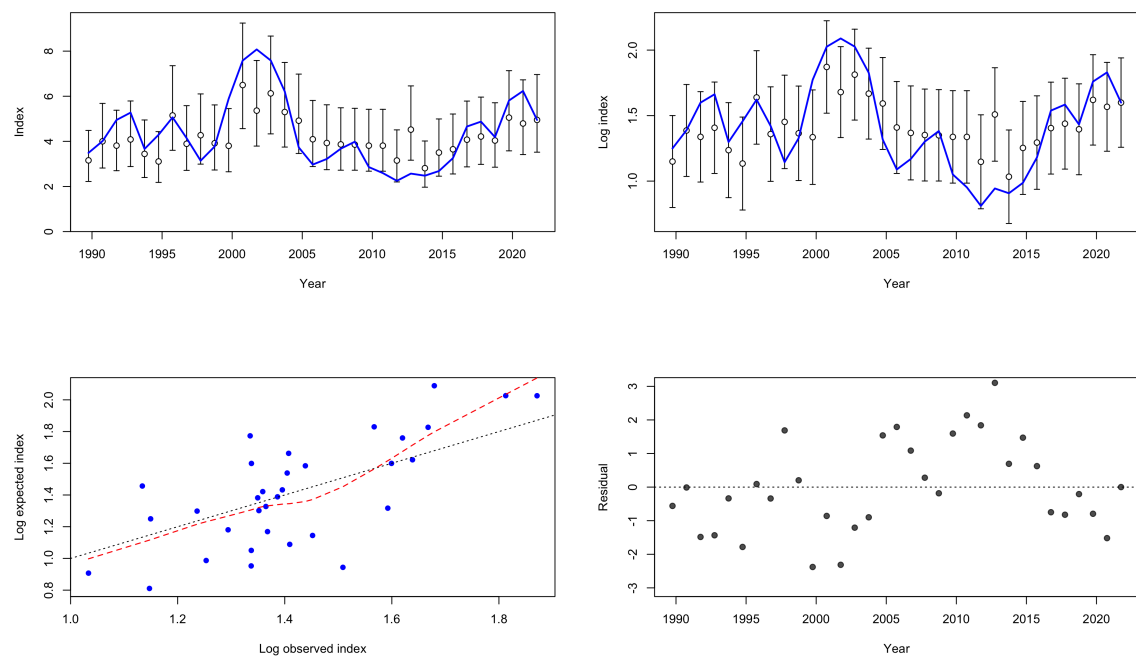


Figure 12: Model fit to South RecCPA index

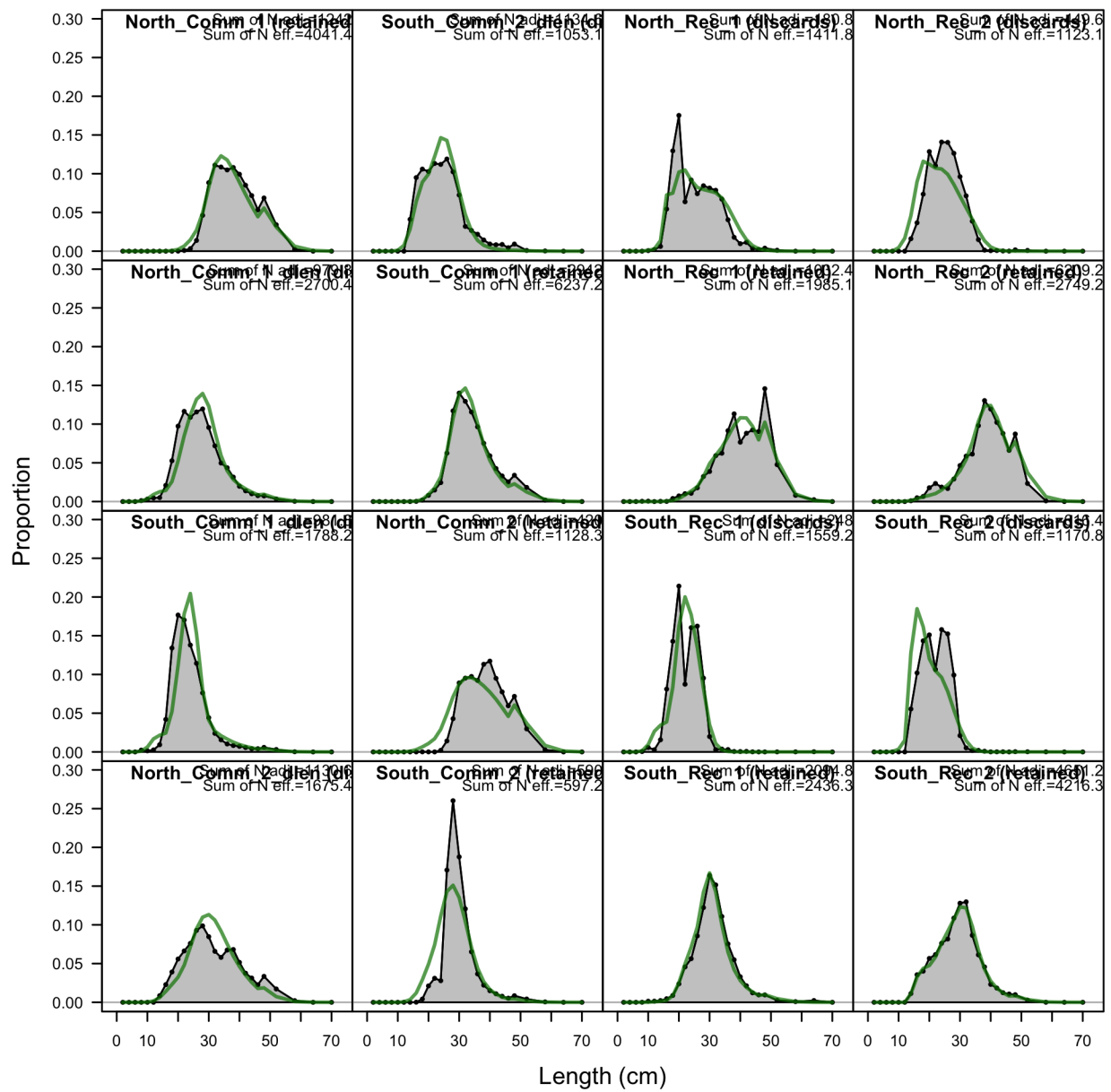
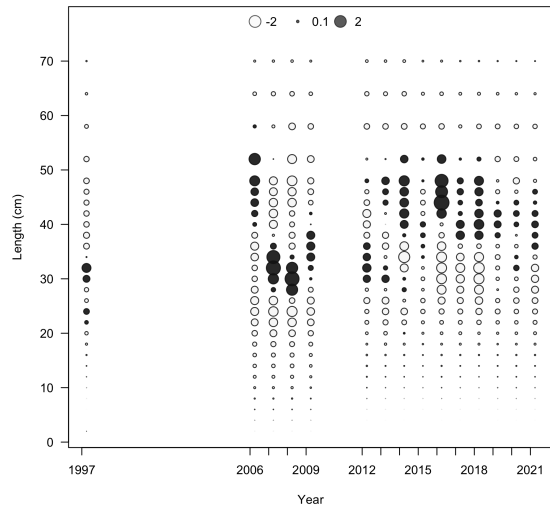
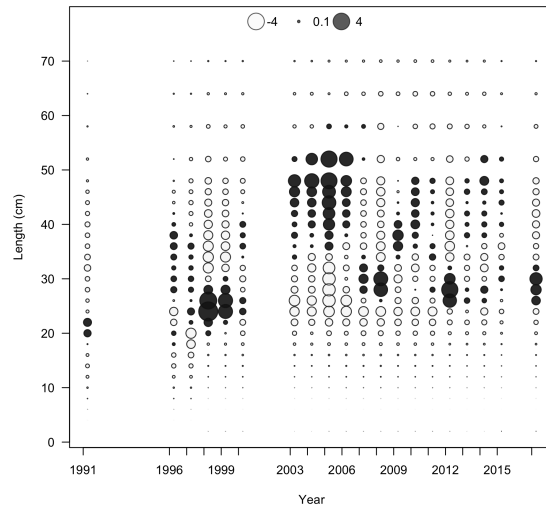


Figure 13: Fits to length composition data, aggregated across time by fleet.

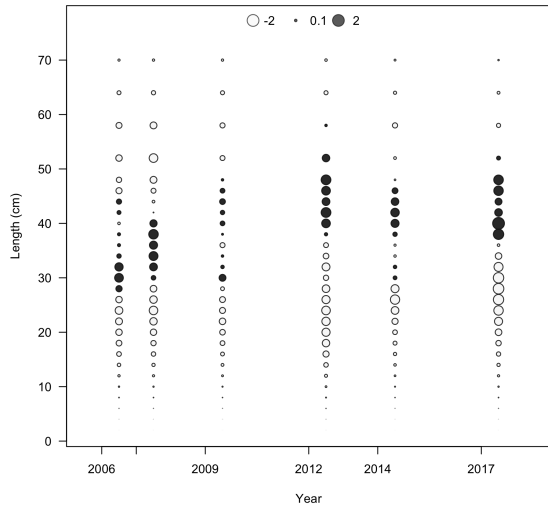
(a) Retained North Commercial Spring



(b) Retained South Commercial Spring



(c) Retained North Commercial Fall



(d) Retained South Commercial Fall

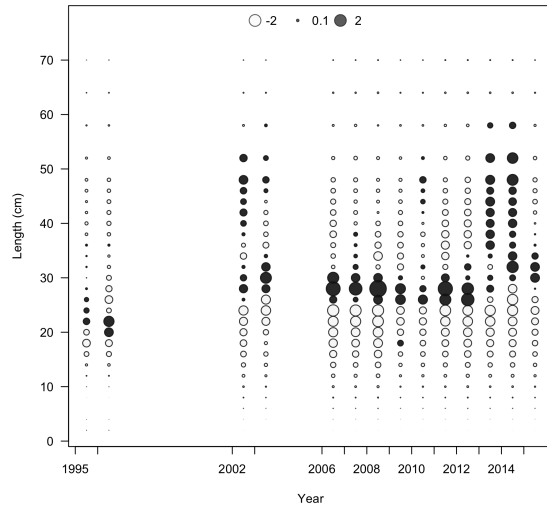
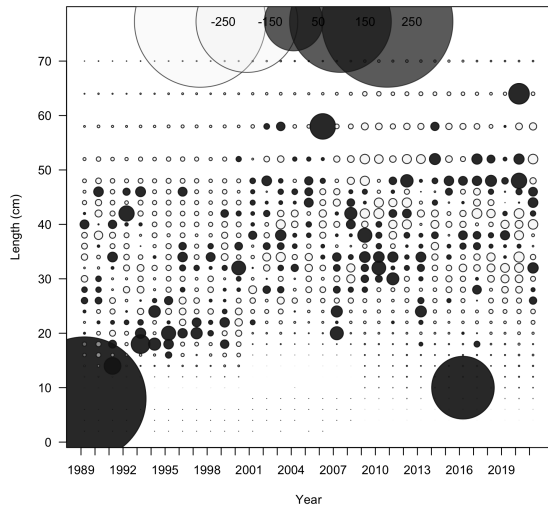
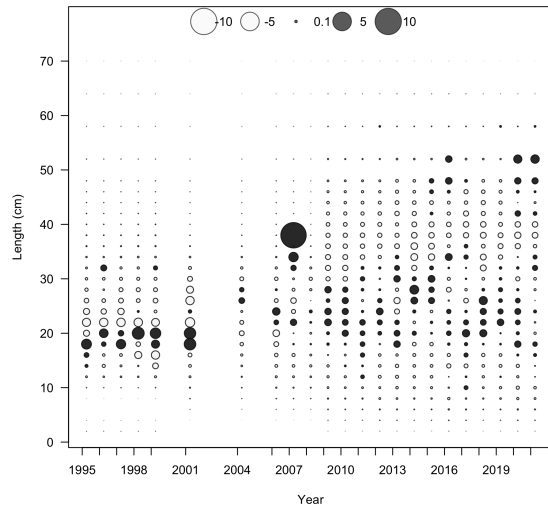


Figure 14: Pearson residuals for fits to commercial retained length composition data

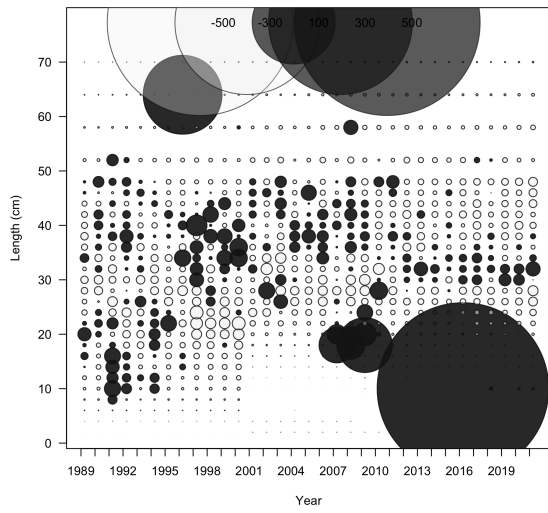
(a) Retained, N Rec Spring



(b) Discards, N Rec Spring



(c) Retained, S Rec Spring



(d) Discards, S Rec Spring

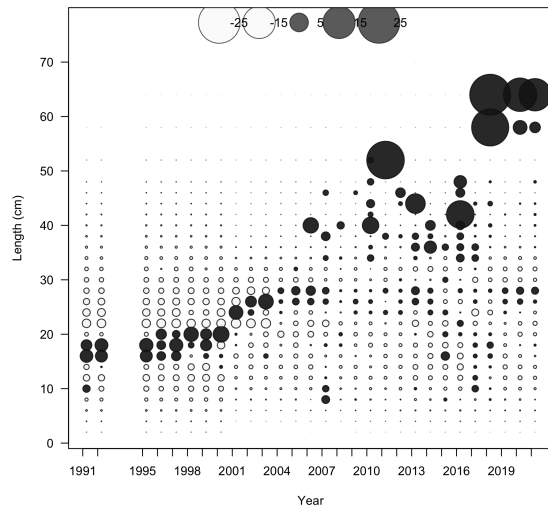
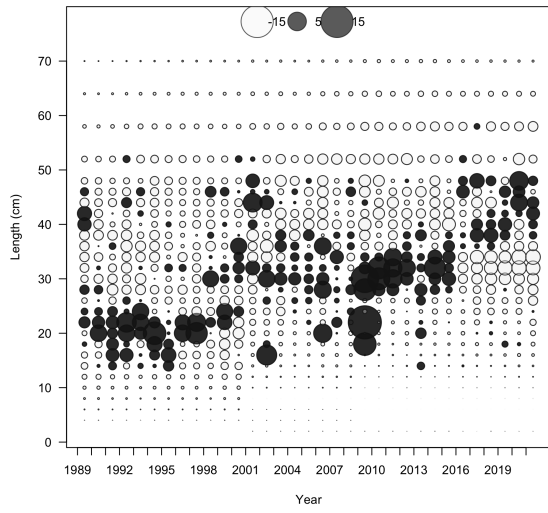
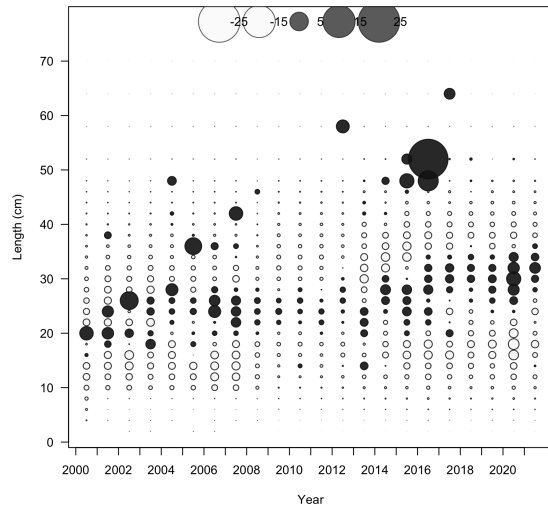


Figure 15: Pearson residuals for fits to spring recreational length composition data

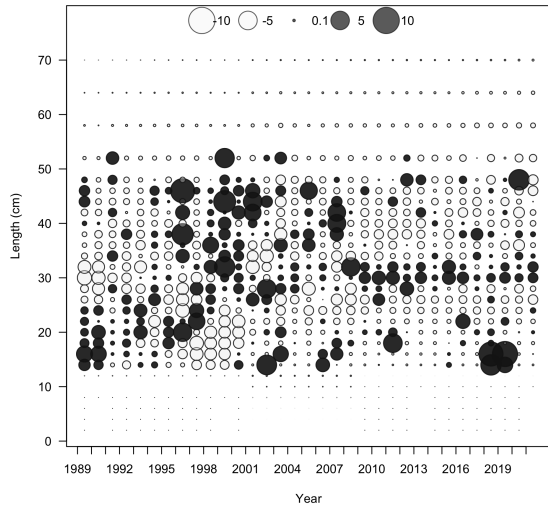
(a) Retained, N Rec Fall



(b) Discards, N Rec Fall



(c) Retained, S Rec Fall



(d) Discards, S Rec Fall

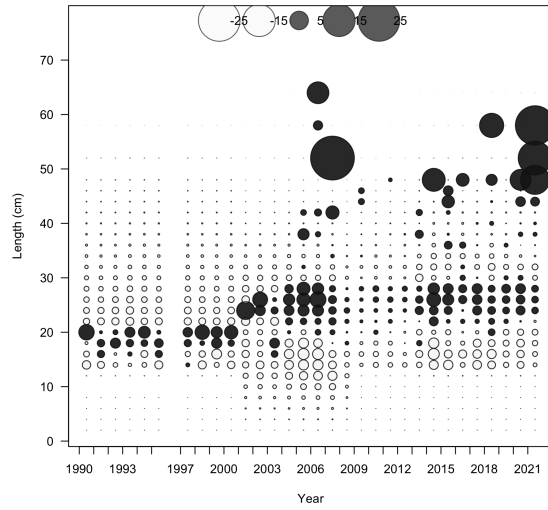
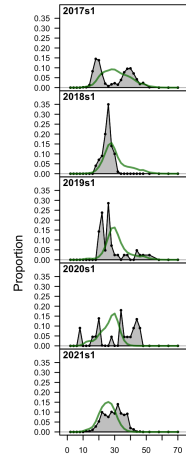


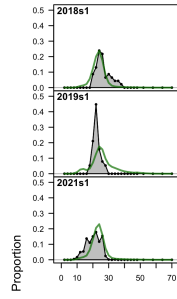
Figure 16: Pearson residuals for fits to fall recreational length composition data

(a) North Commercial Spring



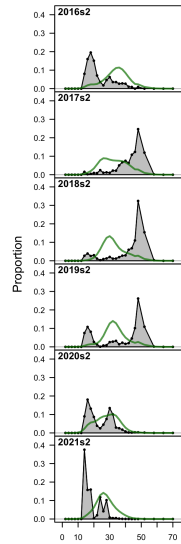
Length (cm)

(b) South Commercial Spring



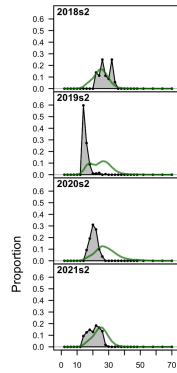
Length (cm)

(c) North Commercial Fall



Length (cm)

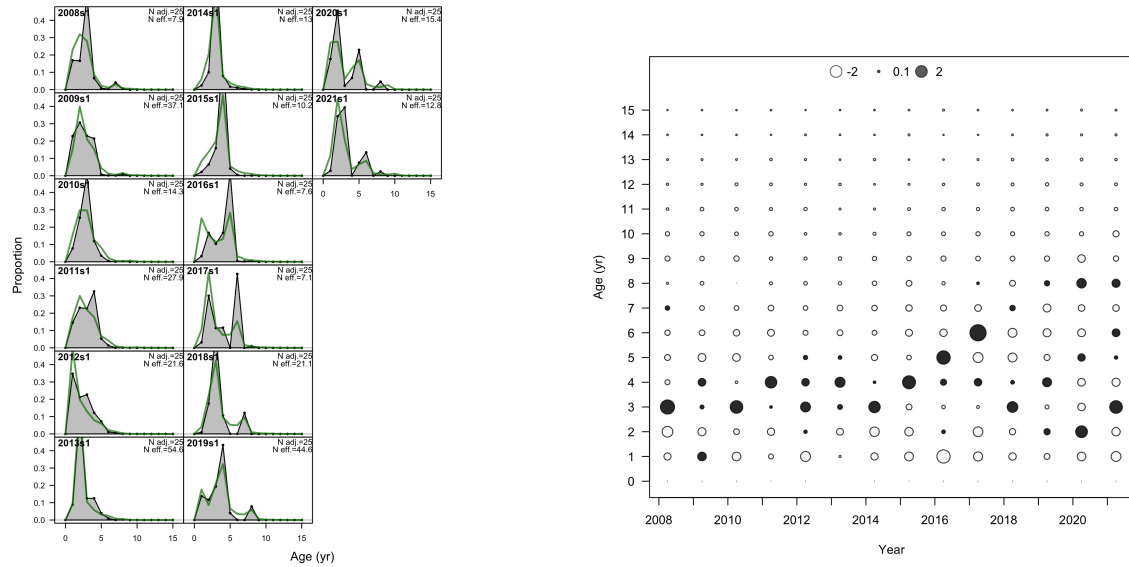
(d) South Commercial Fall



Length (cm)

Figure 17: Implied fits to the CFRF discard data. Shown are the model predictions for these data, though the CFRF discarded length compositions were not included in the likelihood.

(a) VAST North Spring



(b) VAST South Spring

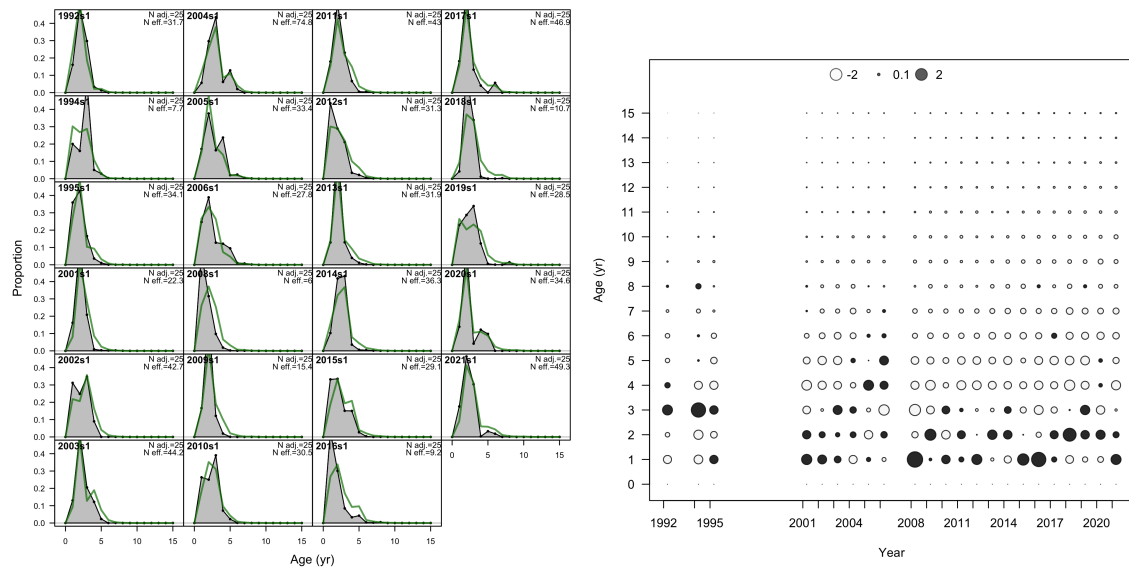


Figure 18: Fits and Pearson residuals to VAST spring survey index age composition for (a) North, and (b) South.

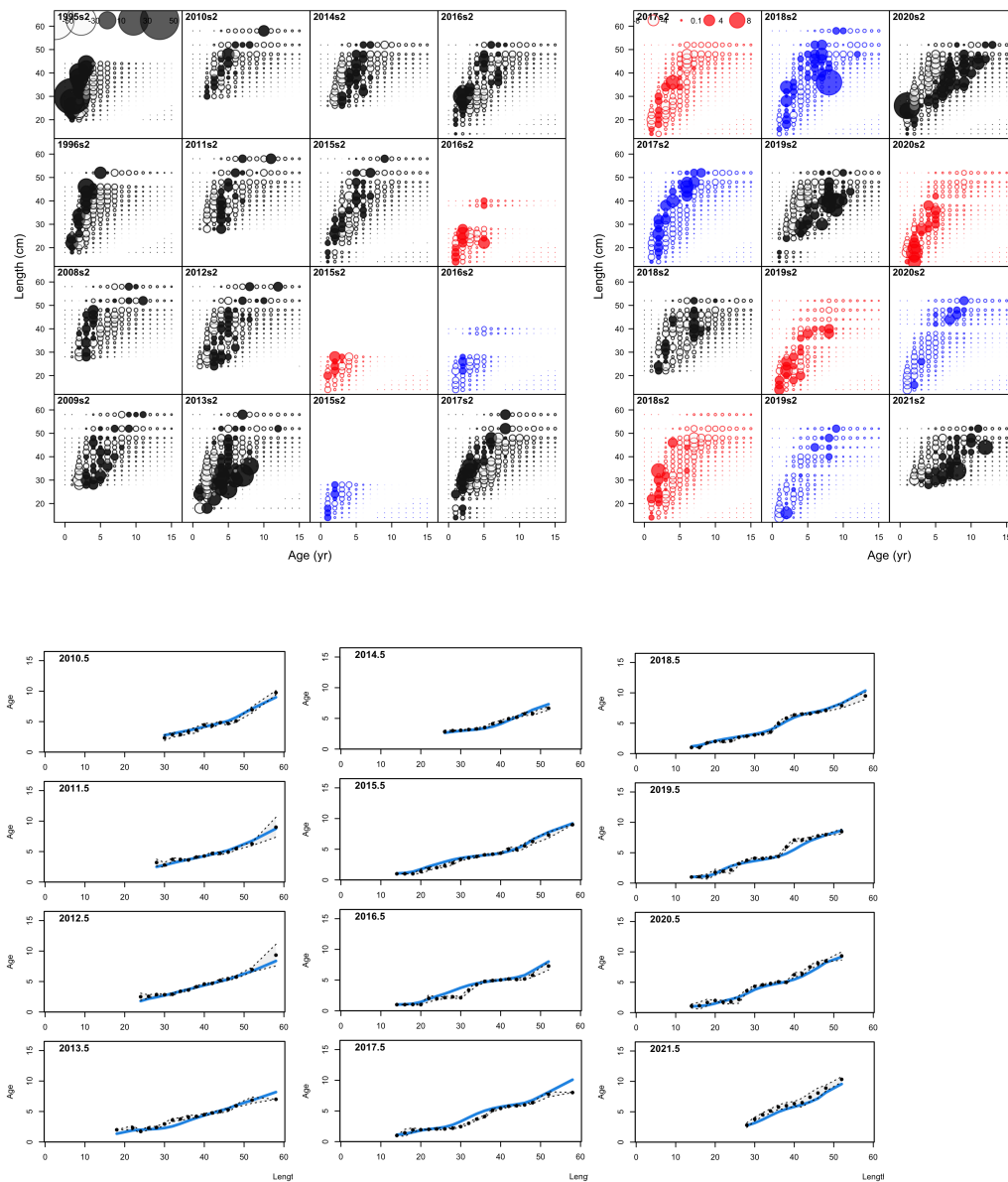


Figure 19: Example pearson residuals and predicted mean age at length for the Fall North Commercial fishery age-at-length data.

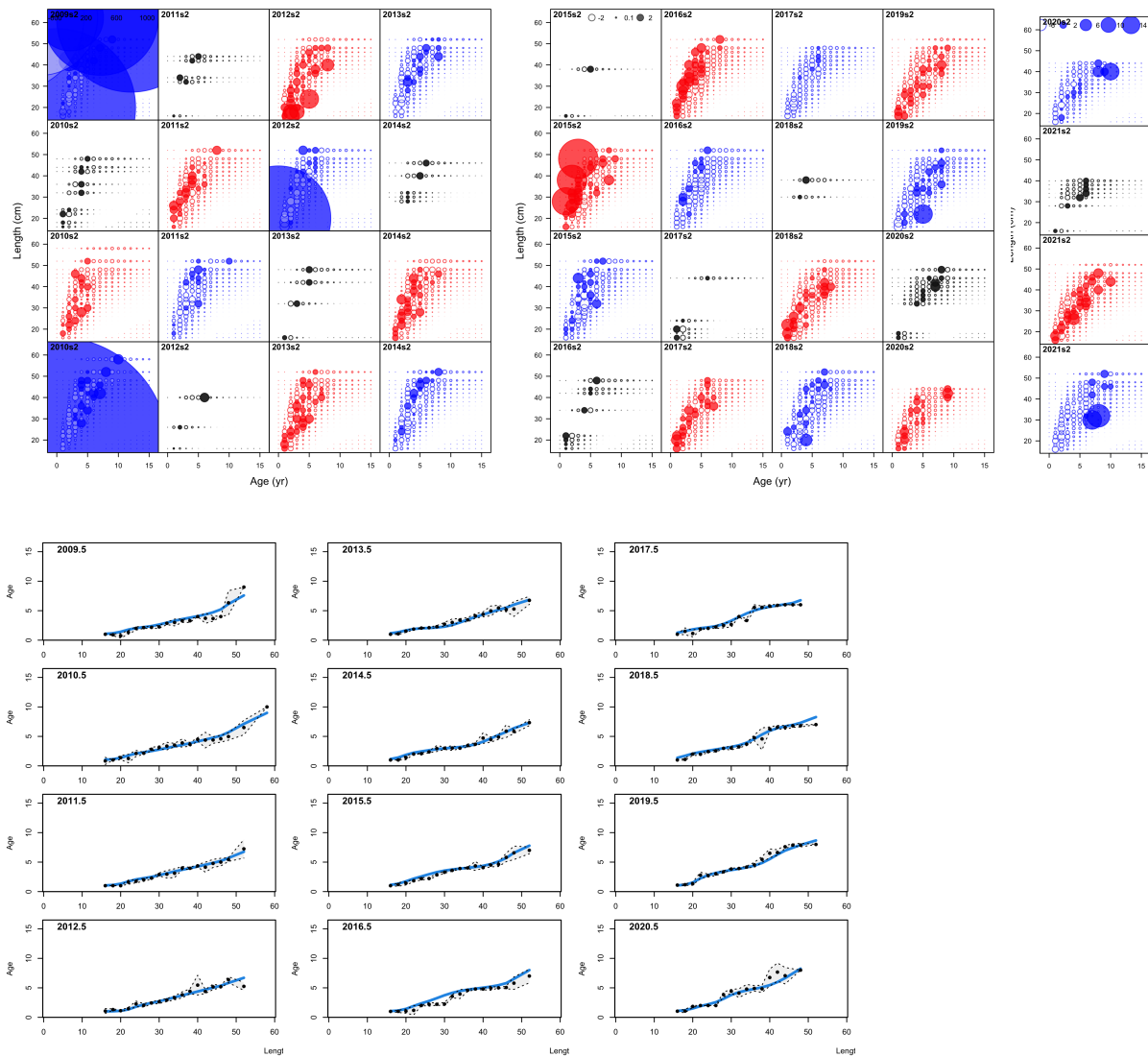


Figure 20: Example pearson residuals and predicted mean age at length for the Fall North survey age-at-length data.

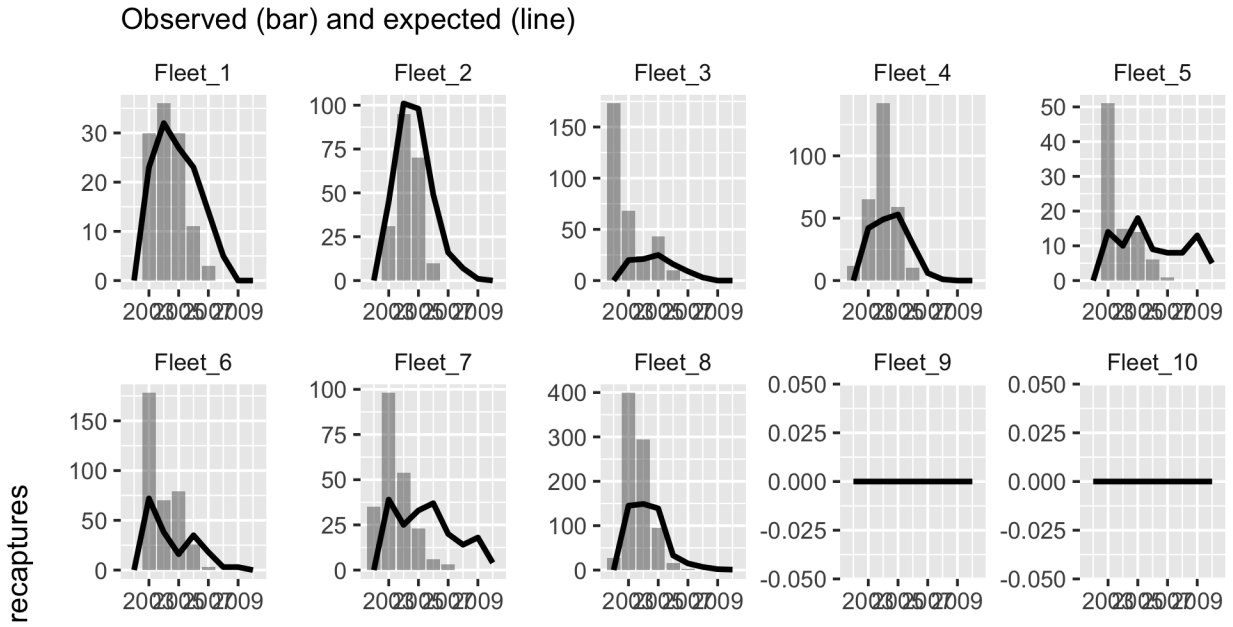
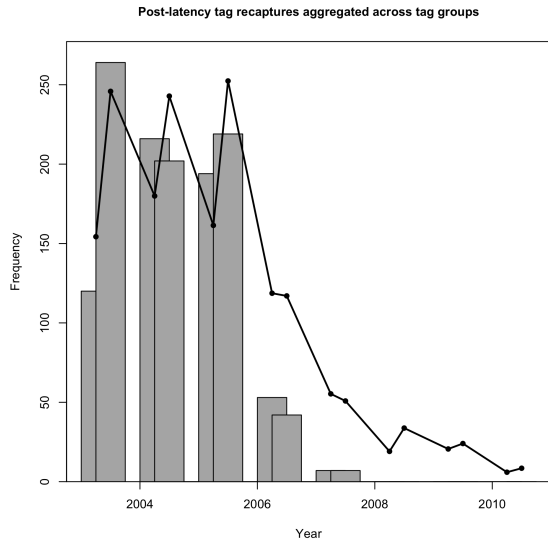


Figure 21: Observed and predicted number of tag recaptures over time, by total and fishing fleet.

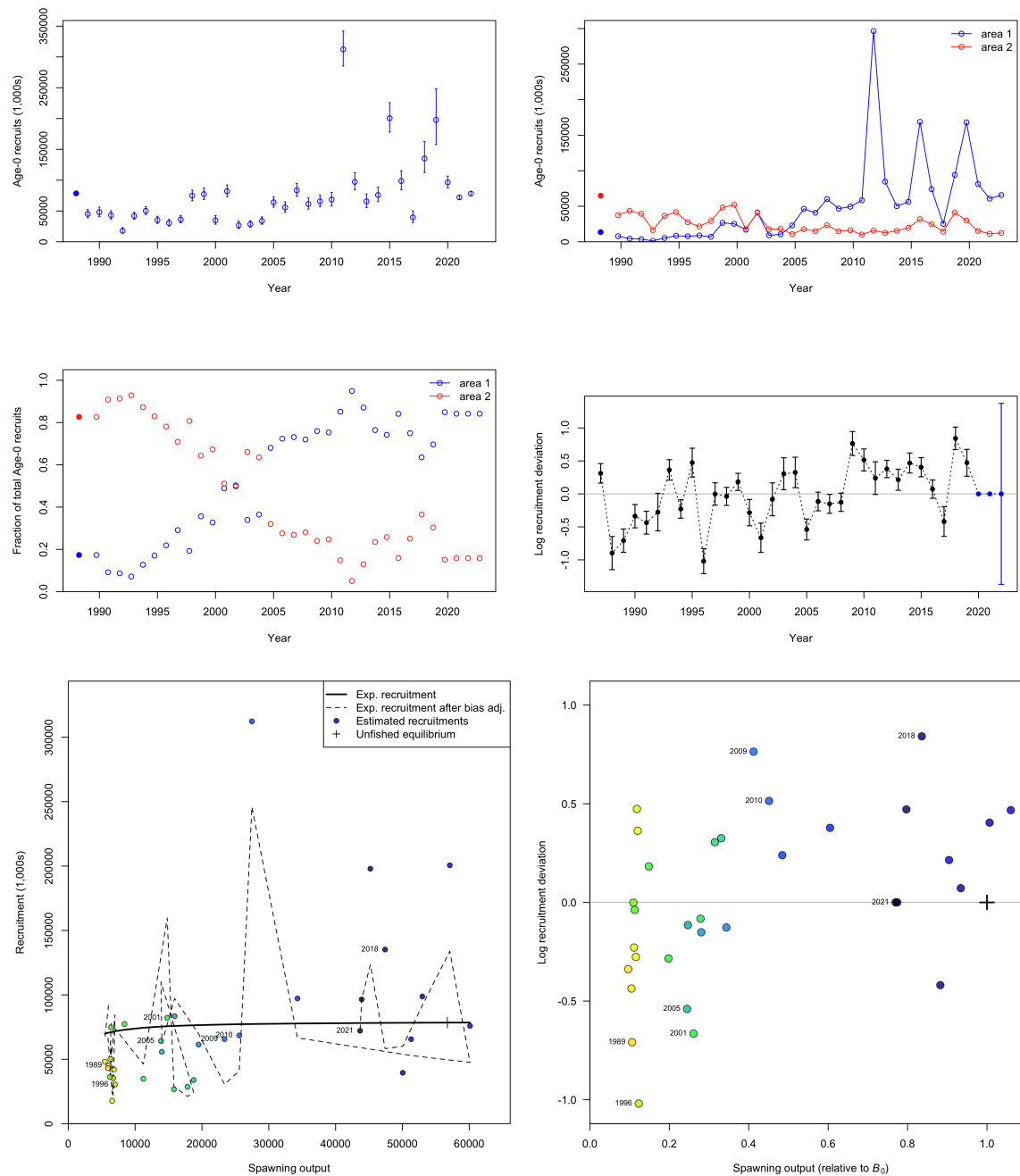


Figure 22: Estimated time series of total recruitment, recruitment by region, proportion of annual recruits allocated to each region, and recruitment deviations, including relative to predicted values from the stock-recruit relationship. Dashed line on bottom left panel corresponds to expected recruitment following bias adjustment and the annual effects of the bottom temperature covariate.

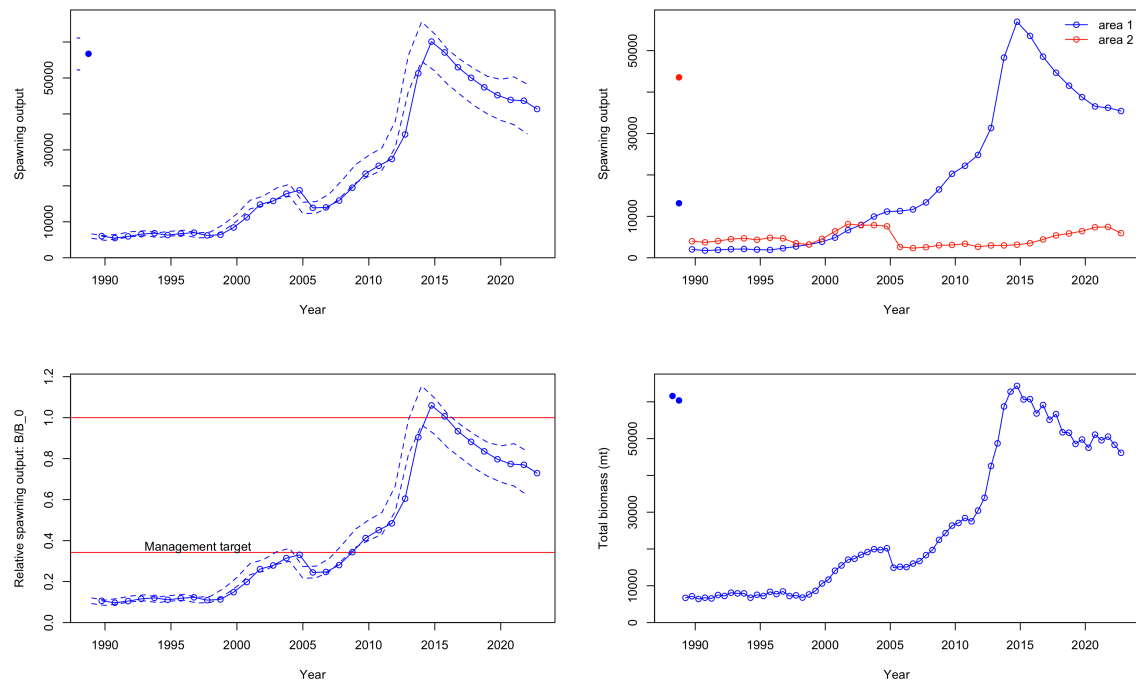


Figure 23: Time series of spawning biomass (both sexes), including total, by area, and relative to management target (B40), and total biomass.

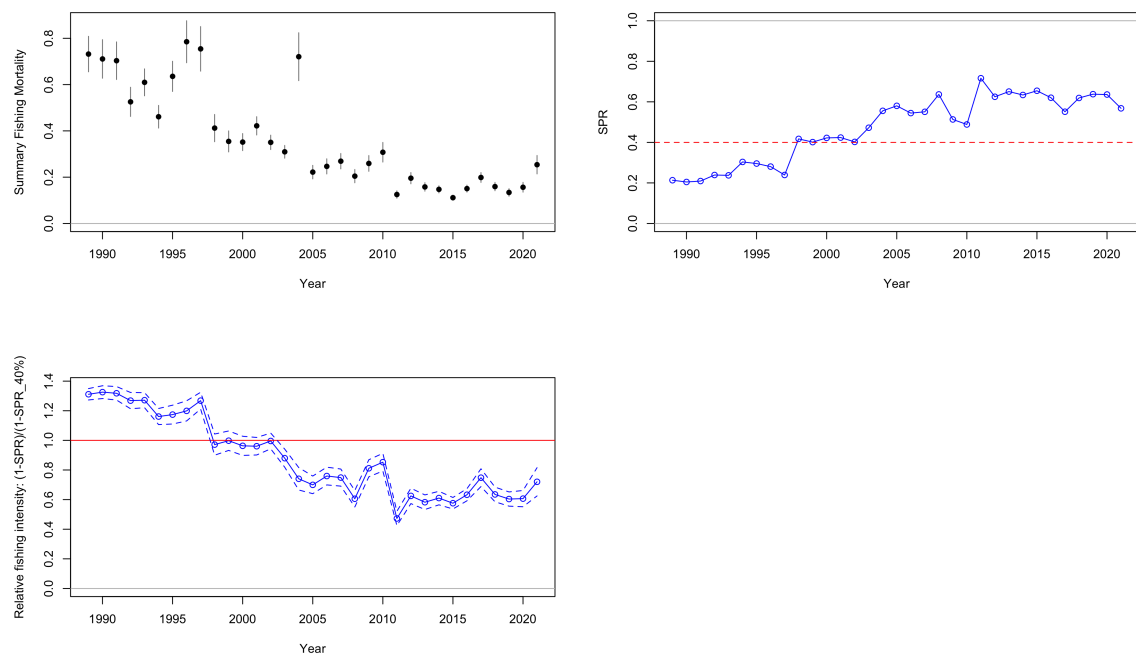


Figure 24: Time series of the estimated values for fishing mortality (summarized as average of F for ages 4-7), SPR, and fishing intensity relative to the $F_{40\%}$ FMSY proxy.

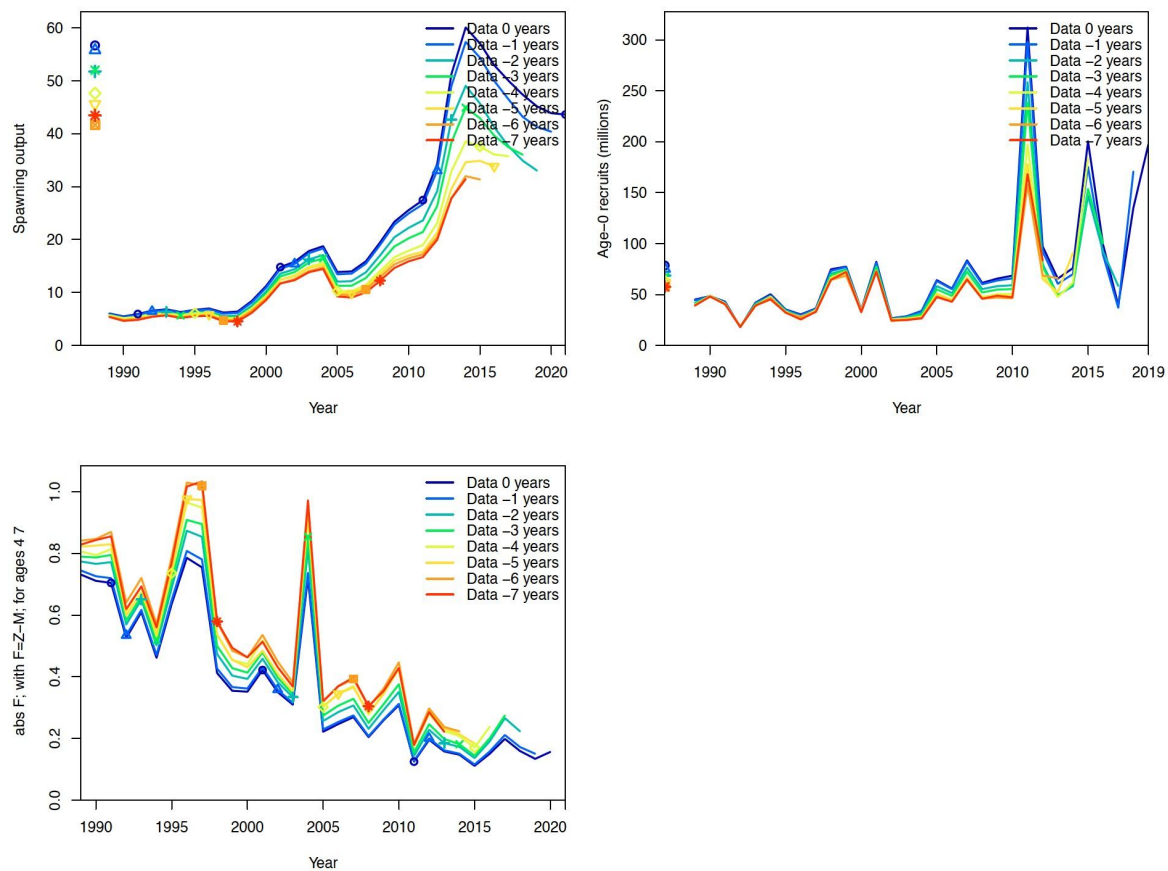


Figure 25: Retrospective analysis results for estimated time series of (a) SSB, (b) recruitment, (c) fishing mortality.

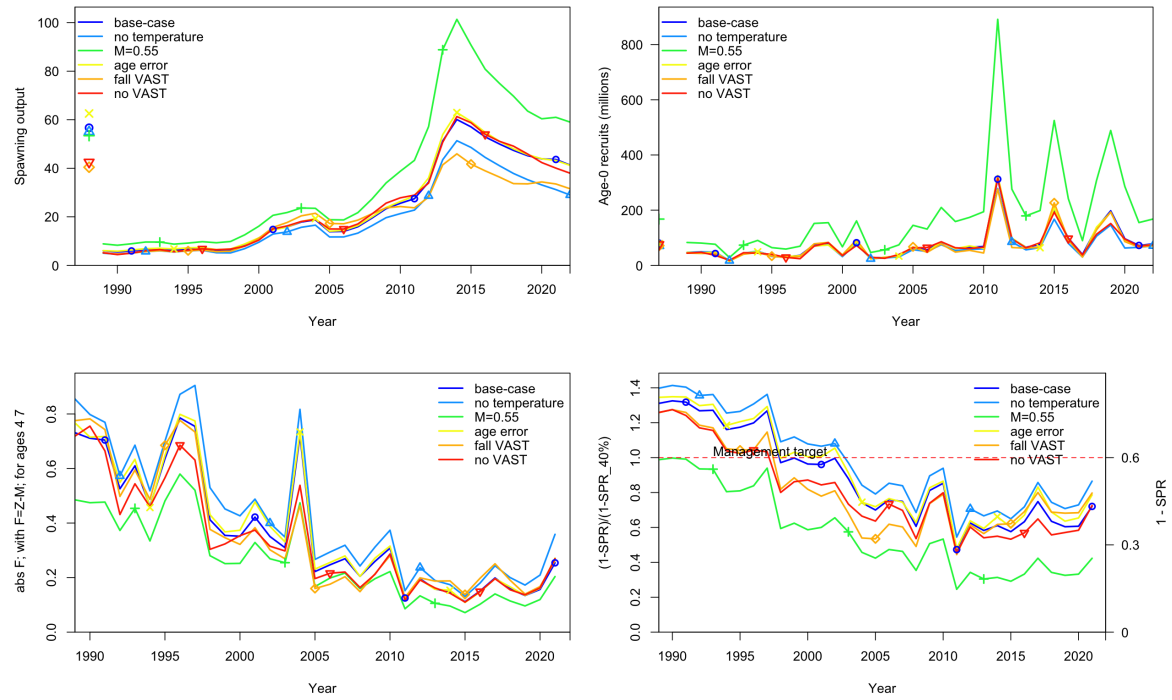


Figure 26. Estimated time series for spawning biomass, recruitment, and fishing mortality for the sensitivity analyses compared to the base-case run.